

Declaration of Generative AI Use

Unit: FIT3143 Parallel Computing, Semester 2 2026

Assessment: Lab 2, Message Passing Interface (Week 8)

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Date: 16 September 2026

We used generative AI while preparing this lab, which item 9 of the assessment specification allows. We did not use any AI tool during the presentation or the Q&A, as item 10 requires.

This is our one AI declaration file. It has three parts:

Part	What it is
Part 1	This declaration
Part 2	The full Gemini Pro prompt record, exported from Gemini
Part 3	The full Claude prompt record, exported from the Claude Code session logs

All measurements in this submission were taken on a MacBook Pro 14 inch (November 2023), Apple M3 Max, 14 cores: 10 performance and 4 efficiency, and on the Monash CAAS cluster.

Tools used

Tool	Used by	Prompt record
Gemini Pro (Google)	Taabish Farooq Bhat and Erwyna Soo Wen Xin, together	Part 2 of this file
Claude (Anthropic)	Erwyna Soo Wen Xin	Part 3 of this file

Gemini Pro

One session, "Hybrid MPI and OpenMP Programming", which Taabish and Erwyna worked through together. The transcript was exported on 16 September 2026. Six prompts, in order:

1. Structuring the hybrid program: `MPI_Init_thread` with `MPI_THREAD_FUNNELED`, an MPI stride across processes, and `#pragma omp parallel for schedule(dynamic, 1000)` inside each process to avoid thread level load imbalance and false sharing.
2. Reviewing our cyclic partitioning, where rank r takes $3 + 2r$ and strides by $2p$, for residue class lock in and the load imbalance when an odd prime d divides p , with $p_{\text{eff}} = p(1 - 1/d)$. Gemini recommended block cyclic chunking with a chunk size of 32,768 odd candidates.
3. Designing the timing framework: `MPI_Wtime` with explicit `MPI_Barrier` calls, so that time a rank spends waiting is not counted as communication, separating the search, the gather (κ) and the root's serial sort and file write (r_s).
4. Benchmark configurations comparing $p \times t$ hybrid runs against pure MPI and pure OpenMP at matched total worker counts, for example 2×7 against 14 processes and against 14 threads.
5. Drafting a 6 to 7 minute presentation script covering methodology, the phase breakdown, the speedup plateau, limitations and multi node CAAS scaling.
6. A practice viva: four questions on load imbalance, residue congruence, shared memory against message passing overheads, and deriving the Amdahl's Law parameters, marked against the HD rubric.

The hybrid structure in `task2.c` follows the first prompt, and the timing method in the instrumented programs follows the third. **The block cyclic suggestion in the second prompt was not adopted.** The submitted `task1.c` and `task2.c` keep the single value stride, and section 11 of `Task3_Performance_Evaluation.pdf` lists that change as future work, so that every measurement we report matches the code we submit.

Claude (Anthropic)

Used by Erwyna on 8, 9, 15 and 16 September 2026.

Task 1 and Task 2 (`task1.c` and `task2.c`, written by Taabish)

- Reviewing the programs. This found one real bug: `task2.c` called `memcpy` without including `<string.h>`, so it did not compile with Apple clang. The include was added.
- Renaming the files to `task1.c` and `task2.c` to match the submission checklist, and rewriting some of the code comments to make them clearer.
- Adding the per rank search times to `task1.c` and the per thread search times to `task2.c`, and adding checks on the value of `n`, on the MPI thread support level, and on every memory allocation. The search, the partitioning and the MPI and OpenMP calls that do the work were not changed.

Task 3 (performance evaluation, written by Erwyna)

- Building the measurement scripts in `supporting/scripts/` and the CAAS job files, and working out why our first CAAS jobs failed to build, until job 39361 ran.
- The second sweep that measures the theoretical speedup as `n` grows (`run_phases_by_n.py`, Graph 10).
- Checking the Amdahl's Law working, and drafting and editing `Task3_Performance_Evaluation.pdf` and `CAAS_Analysis.pdf`.

Task 4 (slides)

- Drafting slide text, drawing the three explanatory diagrams with `make_diagrams.py`, and checking the deck against the specification and the marking rubric.
- On 16 September, a last check of the finished deck and the submission files against the specification, the rubric and our CSV data. This corrected one percentage on appendix slide A5 and in section 7 of the Task 3 document, and put this declaration and both prompt records into one file.

Q&A preparation

- Practice questions to rehearse with before the lab.

Every number in our slides and documents comes from running our own code, on our own laptop and on CAAS, and the scripts in `supporting/scripts/` reproduce them.

Signatures

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We declare that this is a complete and accurate account of our use of generative AI in this assessment, and that we understand and can explain all of the submitted work.

Part 2. AI Prompt Record: Gemini Pro (Google)

Unit: FIT3143 Parallel Computing, Semester 2 2026

Assessment: Lab 2, Message Passing Interface (Week 8)

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The next 7 pages are the full transcript of our one Gemini Pro session for Lab 2, "Hybrid MPI and OpenMP Programming", which Taabish and Erwynna worked through together. It was exported from Gemini on 16 September 2026 and is included unedited: all six prompts and every response. The only thing added is the line with our names along the bottom of each page.

Hybrid MPI and OpenMP Programming

<https://gemini.google.com/app/e94e66f0b5e3168d>

User prompt: Help me structure a hybrid Open MPI + OpenMP program using MPI_Init_thread with MPI_THREAD_FUNNELED. Design a scheme where MPI strides across processes while OpenMP dynamically distributes work inside each process using #pragma omp parallel for schedule(dynamic, 1000) to eliminate thread-level load imbalance and false sharing.

Response: Hybrid Open MPI and OpenMP computing combines distributed-memory coarse partitioning via MPI processes with shared-memory fine-grained multi-threading via OpenMP.

```
#include <mpi.h>
#include <omp.h>
#include <stdio.h>
#include <stdlib.h>

#define N 100000000LL

int main(int argc, char *argv[]) {
    int provided;

    // 1. Initialize MPI with funneled thread support
    MPI_Init_thread(&argc, &argv, MPI_THREAD_FUNNELED, &provided);
    if (provided < MPI_THREAD_FUNNELED) {
        fprintf(stderr, "Error: Required MPI_THREAD_FUNNELED level not supported.\n");
        MPI_Abort(MPI_COMM_WORLD, 1);
    }

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    // 2. Coarse MPI Domain Decomposition (Process-level chunking)
    long long chunk_per_process = N / size;
    long long start_idx = rank * chunk_per_process;
    long long end_idx = (rank == size - 1) ? N : start_idx + chunk_per_process;
    long long local_size = end_idx - start_idx;

    double *local_data = (double *)malloc(local_size * sizeof(double));

    // 3. Intra-process Dynamic OpenMP Scheduling
    #pragma omp parallel for schedule(dynamic, 1000)
    for (long long i = 0; i < local_size; i++) {
        long long global_i = start_idx + i;
        // Example workload computation
        local_data[i] = (double)(global_i * 2);
    }

    // Accumulation workload
    double local_sum = 0.0;
    #pragma omp parallel for schedule(dynamic, 1000) reduction(+:local_sum)
    for (long long i = 0; i < local_size; i++) {
        local_sum += local_data[i];
    }

    // 4. MPI Communication (Executed strictly by the main thread)
    double global_sum = 0.0;
    MPI_Reduce(&local_sum, &global_sum, 1, MPI_DOUBLE, MPI_SUM, 0, MPI_COMM_WORLD);

    if (rank == 0) {
        printf("Computation complete. Global sum: %f\n", global_sum);
    }

    free(local_data);
    MPI_Finalize();
    return 0;
}
```

Key Structural Considerations

- **Thread Support (MPI_THREAD_FUNNELED):** Informs the MPI runtime that while the application is multi-threaded, only the main thread will make MPI calls. All MPI_* routines must reside outside active #pragma omp parallel blocks or inside master-only sections.
- **Coarse vs. Fine Work Distribution:** MPI splits the array across global nodes in contiguous sub-blocks (start_idx to end_idx). OpenMP then distributes each process's local domain across worker threads.
- **Load Imbalance Mitigation:** #pragma omp parallel for schedule(dynamic, 1000) dynamically hands out chunks of 1,000 iterations to worker threads on demand. If a thread finishes early due to variable computation costs, it immediately grabs the next chunk.

- **False Sharing Prevention:** Setting the chunk size to 1000 ensures that worker threads write to large, contiguous byte sequences (8,000 bytes for `double`). Because this significantly exceeds standard 64-byte L1/L2 cache lines, worker threads rarely write to the same cache line, effectively preventing false sharing cache-invalidation cycles.

User prompt: Act as a parallel computing expert. Analyze my Open MPI C implementation for searching primes below \$n\$. Evaluate my cyclic partitioning scheme (\$3 + 2r\$ with stride \$2p\$) for residue class lock-in and load imbalance (e.g., odd prime factors \$d \nmid p\$ where \$p_{\text{eff}} = p(1 - 1/d)\$). Suggest an optimized block-cyclic or interleaved chunking scheme to ensure optimal load distribution across all ranks.

Response: Cyclic striding by $2p$ locks each MPI process into a fixed residue class modulo $2p$, creating severe load imbalance whenever small prime factors divide process count p . To eliminate this imbalance and preserve memory locality, ranks must process continuous block-cyclic chunks that span all residue classes modulo small primes.

Residue Class Lock-in Analysis

- **Modulo Invariance:** Testing odd numbers via $x(r, k) = 3 + 2r + 2pk$ means rank r tests numbers where $x_k \equiv (3 + 2r) \pmod{2p}$. For any odd prime factor $d \mid p$:

$$x(r, k) \equiv (3 + 2r) \pmod{d}$$

- **Workload Disparity:**
 - **Trivial Ranks** ($3 + 2r \equiv 0 \pmod{d}$): Every number assigned to rank r is a multiple of d . Trial division aborts on the very first factor $O(1)$, leaving the rank virtually idle.
 - **Candidate Ranks** ($\gcd(3 + 2r, 2p) = 1$): These ranks process numbers coprime to small factors, requiring heavy trial division up to $O(\sqrt{x})$.
- **Efficiency Decay:** When $d \mid p$, a fraction $1/d$ of ranks are wasted on multiples of d . The proportion of fully active ranks shrinks by the Euler-like factor:

$$p_{\text{eff}} = p \prod_{d \mid p} \left(1 - \frac{1}{d}\right)$$

Partitioning Strategy Comparison

Strategy	Residue Lock-In	Macro Load Balance ($O(\sqrt{x})$ Growth)	Cache Locality
Cyclic ($2p$ Stride)	Severe (High)	Excellent	Poor (strided memory)
Pure Block	None	Poor (higher ranks get larger x)	Excellent
Block-Cyclic Chunking	None	Excellent	Excellent

Optimized Interleaved Block-Cyclic Scheme

Divide odd candidates into contiguous blocks of size B (e.g., $B = 32,768$ elements to fit L1/L2 cache). Distribute global chunk index C to rank r when $C \equiv r \pmod{p}$.

```
// B: Chunk size (number of odd integers per block)
long long B = 32768;
long long total_odds = (n - 3) / 2 + 1;
long long total_chunks = (total_odds + B - 1) / B;

for (long long C = rank; C < total_chunks; C += size) {
    long long start_idx = C * B;
    long long end_idx = (start_idx + B < total_odds) ? start_idx + B : total_odds;

    // Process contiguous odds in chunk C
    #pragma omp parallel for schedule(dynamic, 1000)
    for (long long i = start_idx; i < end_idx; i++) {
        long long val = 3 + 2 * i;
        if (is_prime(val)) {
            // Count or collect prime
        }
    }
}
```

Why Block-Cyclic Resolves the Bottlenecks

- **Uniform Residue Coverage:** Each block B sweeps continuously across consecutive odds, ensuring every rank handles an equal distribution of numbers congruent to $0 \pmod{3}$, $0 \pmod{5}$, and coprime candidates.

- **Asymptotic Load Balancing:** Interleaving blocks across ranks ensures that no single rank is trapped with only the largest candidate numbers where x trial division costs are highest.

User prompt: Design an experimental timing framework using `MPI_Wtime()` and explicit `MPI_Barrier()` calls to isolate search time, gather communication time ($\kappa(p)$), and root serial processing time (r_s via `qsort` and file writing) without skewing idle rank waiting times into communication costs.

```
Response: #include <mpi.h>
#include <stdio.h>
#include <stdlib.h>

int compare_long_long(const void *a, const void *b) {
    long long arg1 = *(const long long *)a;
    long long arg2 = *(const long long *)b;
    if (arg1 < arg2) return -1;
    if (arg1 > arg2) return 1;
    return 0;
}

void run_benchmarked_search(int rank, int size, long long *local_primes, int local_count) {
    double t_start_search, t_end_search;
    double t_start_comm, t_end_comm;
    double t_start_root, t_end_root;

    // -----
    // Phase 1: Pure Search Computation
    // -----
    t_start_search = MPI_Wtime();

    // [ Prime search execution occurs here ]

    t_end_search = MPI_Wtime();
    double t_search_local = t_end_search - t_start_search;

    // -----
    // Synchronize to isolate load imbalance idle time from communication costs
    // -----
    MPI_Barrier(MPI_COMM_WORLD);

    // -----
    // Phase 2: Gather Communication Time ( $\kappa(p)$ )
    // -----
    int *recvcounts = NULL;
    int *displs = NULL;
    long long *global_primes = NULL;
    int total_primes = 0;

    t_start_comm = MPI_Wtime();

    if (rank == 0) {
        recvcounts = (int *)malloc(size * sizeof(int));
    }

    // Gather sizes from all processes
    MPI_Gather(&local_count, 1, MPI_INT, recvcounts, 1, MPI_INT, 0, MPI_COMM_WORLD);

    if (rank == 0) {
        displs = (int *)malloc(size * sizeof(int));
        displs[0] = 0;
        for (int i = 1; i < size; i++) {
            displs[i] = displs[i - 1] + recvcounts[i - 1];
        }
        total_primes = displs[size - 1] + recvcounts[size - 1];
        global_primes = (long long *)malloc(total_primes * sizeof(long long));
    }

    // Variable gather of results
    MPI_Gatherv(local_primes, local_count, MPI_LONG_LONG,
                global_primes, recvcounts, displs, MPI_LONG_LONG,
                0, MPI_COMM_WORLD);

    t_end_comm = MPI_Wtime();
    double t_comm_local = t_end_comm - t_start_comm;

    // -----
    // Phase 3: Root Serial Processing Time ( $r_s$ )
    // -----
    double t_root_serial = 0.0;
    if (rank == 0) {
```

```

t_start_root = MPI_Wtime();

// 1. Sort combined output
qsort(global_primes, total_primes, sizeof(long long), compare_long_long);

// 2. Sequential file write
FILE *fp = fopen("primes.txt", "w");
if (fp) {
    for (int i = 0; i < total_primes; i++) {
        fprintf(fp, "%lld\n", global_primes[i]);
    }
    fclose(fp);
}

t_end_root = MPI_Wtime();
t_root_serial = t_end_root - t_start_root;

free(recvcounts);
free(displs);
free(global_primes);
}

// -----
// Metrics Aggregation across Ranks
// -----
double t_search_max, t_comm_max, t_search_min;

MPI_Reduce(&t_search_local, &t_search_max, 1, MPI_DOUBLE, MPI_MAX, 0, MPI_COMM_WORLD);
MPI_Reduce(&t_search_local, &t_search_min, 1, MPI_DOUBLE, MPI_MIN, 0, MPI_COMM_WORLD);
MPI_Reduce(&t_comm_local, &t_comm_max, 1, MPI_DOUBLE, MPI_MAX, 0, MPI_COMM_WORLD);

if (rank == 0) {
    double total_wall_time = t_search_max + t_comm_max + t_root_serial;
    double imbalance_ratio = (t_search_max - t_search_min) / t_search_max;

    printf("--- Benchmarking Results (Rank 0) ---\n");
    printf("Search Time (Max): %.6f s\n", t_search_max);
    printf("Comm Time κ(p): %.6f s\n", t_comm_max);
    printf("Root Serial r_s: %.6f s\n", t_root_serial);
    printf("Total Wall Time: %.6f s\n", total_wall_time);
    printf("Search Load Skew: %.2f%%\n", imbalance_ratio * 100.0);
}
}

```

Timing Framework Architecture

- **Barrier Synchronization Placement:** Placing `MPI_Barrier()` after $t_{\text{end_search}}$ forces fast ranks to wait until all processes complete local trial division. This ensures that the time spent waiting for the slowest rank (load imbalance) is absorbed before $t_{\text{start_comm}}$, preventing idle waiting latency from inflating the gather timing $\kappa(p)$.
- **Communication Isolation ($\kappa(p)$):** t_{comm} captures only `MPI_Gather` setup, `MPI_Gatherv` collective data transfers, and root-side buffer allocations. Using `MPI_Reduce(..., MPI_MAX)` captures the critical-path bottleneck across all nodes.
- **Root Serial Isolation (r_s):** $t_{\text{root_serial}}$ is executed solely on rank 0 inside a dedicated timing block. It accurately tracks non-parallelizable bottlenecks ($O(N \log N)$ `qsort` and sequential disk I/O) without affecting remote rank timers.
- **Load Imbalance Metric:** Computing $\frac{t_{\text{search_max}} - t_{\text{search_min}}}{t_{\text{search_max}}}$ quantifies worker starvation without contaminating network metrics.

User prompt: Generate C code logic and benchmark configurations comparing $\$p$ times $t\$$ hybrid setups against pure MPI ($\$p\$$ processes) and pure OpenMP ($\$t\$$ threads) at matched total worker counts (e.g., $\$2$ times $7\$$ vs $\$14\$$ processes vs $\$14\$$ threads).

```

Response: #include <mpi.h>
#include <omp.h>
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

bool is_prime(long long n) {
    if (n < 2) return false;
    if (n == 2 || n == 3) return true;
    if (n % 2 == 0 || n % 3 == 0) return false;
    for (long long i = 5; i * i <= n; i += 6) {
        if (n % i == 0 || n % (i + 2) == 0) return false;
    }
    return true;
}

```

```

int main(int argc, char *argv[]) {
    int provided;
    MPI_Init_thread(&argc, &argv, MPI_THREAD_FUNNELED, &provided);

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    long long n = (argc > 1) ? atoll(argv[1]) : 100000000LL;
    int num_threads = omp_get_max_threads();

    // Synchronize processes before clocking
    MPI_Barrier(MPI_COMM_WORLD);
    double t_start = MPI_Wtime();

    long long B = 32768; // Chunk size for L1/L2 locality
    long long total_odds = (n - 3) / 2 + 1;
    long long total_chunks = (total_odds + B - 1) / B;
    long long local_primes = 0;

    // Outer MPI block-cyclic stride across chunks
    for (long long C = rank; C < total_chunks; C += size) {
        long long start_idx = C * B;
        long long end_idx = (start_idx + B < total_odds) ? start_idx + B : total_odds;

        // Inner OpenMP dynamic thread dispatch
        #pragma omp parallel for schedule(dynamic, 1000) reduction(+:local_primes)
        for (long long i = start_idx; i < end_idx; i++) {
            long long val = 3 + 2 * i;
            if (is_prime(val)) {
                local_primes++;
            }
        }
    }

    MPI_Barrier(MPI_COMM_WORLD);
    double t_end = MPI_Wtime();
    double local_time = t_end - t_start;

    double max_time;
    long long global_primes = 0;

    MPI_Reduce(&local_time, &max_time, 1, MPI_DOUBLE, MPI_MAX, 0, MPI_COMM_WORLD);
    MPI_Reduce(&local_primes, &global_primes, 1, MPI_LONG_LONG, MPI_SUM, 0, MPI_COMM_WORLD);

    if (rank == 0) {
        // Output format: Processes, Threads, Total Workers, Total Primes, Max Time (s)
        printf("%d,%d,%d,%lld,%.6f\n", size, num_threads, size * num_threads, global_primes, max_time);
    }

    MPI_Finalize();
    return 0;
}

```

Benchmark Configuration Matrix ($W = 14$ Workers)

Paradigm	MPI Ranks (p)	OMP Threads (t)	Total Workers (W)	Primary Bottleneck Risk
Pure MPI	14	1	14	Memory overhead, MPI process context switching
Pure OpenMP	1	14	14	NUMA cross-socket latency, lock contention
Balanced Hybrid	2	7	14	Thread launch overhead across sockets
Asymmetric Hybrid	7	2	14	Over-decomposed MPI domains vs thread pool size

Execution Commands & Thread Binding

Export environment settings and run the binary using strict hardware locality controls to prevent socket migration during tests:

- **Pure MPI Configuration (14×1):**

```

export OMP_NUM_THREADS=1
mpirun -np 14 --bind-to core ./prime_bench 100000000

```

- **Pure OpenMP Configuration (1×14):**

```

export OMP_NUM_THREADS=14
export OMP_PROC_BIND=true

```



```
export OMP_PLACES=cores
mpirun -np 1 --bind-to socket ./prime_bench 100000000
```

- **Hybrid Configuration (2×7):**

```
export OMP_NUM_THREADS=7
export OMP_PROC_BIND=close
export OMP_PLACES=cores
mpirun -np 2 --map-by numa --bind-to numa ./prime_bench 100000000
```

Comparative Evaluation Metrics

- **NUMA Domain Overhead:** Map 1 MPI process per NUMA socket to restrict thread migration while eliminating remote socket memory access penalties inherent in Pure OpenMP setups.
- **Memory Footprint:** Monitor per-node memory utilization; Pure MPI duplicates communicator buffers and process structures across 14 ranks, whereas Hybrid shares process buffers among 7 threads.
- **Collective Scaling ($\kappa(p)$):** Track communication phase latencies; Hybrid setups reduce the participant process count in collective calls from 14 down to 2 ranks.

User prompt: Draft a concise 6-to-7-minute presentation script covering methodology, phase breakdowns, speedup plateaus, limitations (e.g., single-node shared memory constraints), and future multi-node CAAS cluster scaling.

Response: [0:00 – 1:30] Slide 1: Introduction & Hybrid Methodology

Good morning/afternoon, everyone. Today, I'm presenting our hybrid parallel computing framework designed for high-performance prime candidate searching, combining coarse-grain Open MPI process partitioning with fine-grain OpenMP dynamic multi-threading.

Standard cyclic domain partitioning—like striding by $2p$ across odd numbers—suffers from severe residue-class lock-in. When the process count p shares prime factors like 3 or 5, specific ranks get stuck exclusively processing numbers congruent to those multiples. These "trivial" ranks abort trial division almost instantly in $O(1)$ time, while non-trivial ranks bear the entire computational burden of deep trial division up to $O(x)$. To solve this, we implemented an interleaved block-cyclic chunking scheme. By processing contiguous blocks of $B = 32,768$ elements, every rank sweeps across all residue classes uniformly, completely eliminating structural load skew.

At the intra-process level, we initialize MPI using `MPI_THREAD_FUNNELED`, restricting all network operations to the main thread. Inside each process, we dispatch work using `#pragma omp parallel for schedule(dynamic, 1000)`. Assigning dynamic chunks of 1,000 iterations achieves two crucial goals: it balances varying trial-division times across thread pools, and its 8,000-byte allocation size vastly exceeds standard 64-byte L1/L2 cache lines, completely eliminating false sharing.

[1:30 – 3:00] Slide 2: Execution Phase Breakdown & Benchmark Isolation

To rigorously analyze where execution time is spent, our experimental timing framework uses explicit `MPI_Barrier()` syncs and `MPI_Wtime()` calls to decouple computation, communication, and serial overheads into three isolated phases:

1. **Parallel Search Phase (t_{search}):** Ranks independently perform localized prime searches using our block-cyclic distribution.
2. **Gather Communication Phase ($\kappa(p)$):** An explicit `MPI_Barrier()` forces fast processes to wait for the slowest worker *before* initiating `MPI_Gatherv`. This guarantees that idle times from residual computational load skews are fully absorbed into search metrics and never wrongly misclassified as network latency.
3. **Root Serial Processing Phase (r_s):** Rank 0 performs sequential $O(N \log N)$ sorting via `qsort` and writes output results to disk.

By tracking $t_{\text{search_max}}$, $t_{\text{comm_max}}$, and r_s independently, we can quantify exact communication overheads and load imbalance ratios without metric contamination.

[3:00 – 4:30] Slide 3: Speedup Plateaus & Hardware Bottlenecks

When comparing matched-worker configurations—such as 14 Pure MPI processes vs. 14 Pure OpenMP threads vs. a 2×7 Hybrid configuration—we observe distinct scaling plateaus governed by hardware boundaries and Amdahl's Law.

- **Pure OpenMP (1×14)** suffers at high core counts due to NUMA cross-socket memory latencies and cache coherency traffic across CPU socket interconnects.
- **Pure MPI (14×1)** delivers strong search performance but hits memory footprint constraints and higher collective aggregation overhead during `MPI_Gatherv` operations.
- **Hybrid (2×7)** balances these trade-offs by pinning 1 MPI process per NUMA domain and distributing 7 OpenMP threads locally within each socket's shared L3 cache.

However, as worker counts scale, overall speedup plateaus. This is driven primarily by Amdahl's Law: the serial processing phase r_s on Rank 0—specifically single-threaded `qsort` and sequential disk I/O—becomes the dominant bottleneck as parallel compute time drops toward zero.

[4:30 – 6:30] Slide 4: Limitations & Future CaaS Cluster Scaling

Our current architecture faces a core single-node shared-memory limitation: scaling worker counts within a single machine eventually saturates memory bus bandwidth and bus request queues.

To break past these single-node limits, our target future architecture involves scaling to multi-node CaaS (Container-as-a-Service) HPC clusters. In this model, containerized workloads will be deployed across independent physical nodes, mapping 1 MPI process per container/NUMA node with local OpenMP thread pinning.

To ensure true linear scaling in multi-node environments, we will upgrade our framework to resolve the serial r_s bottleneck:

- Replace single-node `qsort` with a distributed parallel merge-sort across all MPI ranks.
- Replace sequential file writes with collective `MPI-IO` (`MPI_File_write_at_all`), allowing ranks to stream prime output concurrently directly into shared persistent storage.

Thank you. I am now open to any questions.

User prompt: Act as a Monash University lab assessor for FIT3143. Ask me 4 challenging Q&A questions probing load imbalance, residue congruence, shared-memory vs message-passing overheads, and Amdahl's law parameter derivations. Evaluate my responses strictly against the HD rubric for correct use of parallel computing terminology (e.g., inter-process communication, lock contention, thread funneled, cache line invalidation, core oversubscription).

Response: Welcome to your FIT3143 viva assessment. To achieve a High Distinction (HD) mark, your answers must demonstrate rigorous mathematical derivations, precise architectural reasoning, and exact parallel computing terminology—including concepts like inter-process communication, lock contention, thread funneled runtime semantics, cache line invalidation, NUMA domain affinity, and core oversubscription.

Answer the following four assessment questions:

1. Residue Congruence & Load Imbalance Mathematically prove why cyclic partitioning ($3 + 2r$ with stride $2p$) causes residue class lock-in whenever an odd prime factor d divides process count p . Explain how this causes asymptotic complexity degradation between trivial ranks ($O(1)$) and candidate ranks ($O(x)$), and demonstrate why interleaved block-cyclic chunking ($B = 32, 768$) mathematically resolves this skew.

2. Shared-Memory vs. Message-Passing Architectural Overheads Compare three matched-worker configurations ($W = 14$): Pure MPI (14×1), Pure OpenMP (1×14), and Hybrid (2×7). Evaluate each against:

- Inter-process communication memory footprint vs. shared-memory thread synchronization overhead.
- NUMA cross-socket memory access latency and cache line invalidation.
- The risk of core oversubscription and thread migration when binding worker threads using `OMP_PROC_BIND` and `OMP_PLACES`.

3. Amdahl's Law Parameter Derivation Given measured parallel search time $t_{\text{search}}(p)$, collective communication time $\kappa(p)$, and root serial processing time r_s (sorting and I/O):

- Derive the formula for total speedup $S(p)$ relative to single-core execution $T(1)$.
- Derive the implicit serial fraction f as a function of r_s , $t_{\text{search}}(1)$, and $\kappa(p)$.
- Calculate the theoretical maximum speedup limit $S_{\text{max}} = \lim_{p \rightarrow \infty} S(p)$ and explain how distributed `MPI-IO` modifies this upper bound.

4. Thread Runtime Semantics & OpenMP Scheduling Mechanics Explain the operational differences between `MPI_THREAD_FUNNELED`, `MPI_THREAD_SERIALIZED`, and `MPI_THREAD_MULTIPLE`. What specific runtime errors or performance penalties occur if an MPI collective call is executed within an un-gated `#pragma omp parallel` region under `MPI_THREAD_FUNNELED`? Furthermore, explain why `#pragma omp parallel for schedule(dynamic, 1000)` eliminates both thread-level load starvation and false sharing at the L1/L2 cache line level.

Please provide your complete answers to all four questions when you are ready to begin your evaluation.

Part 3. AI Prompt Records: Claude (Anthropic)

Unit: FIT3143 Parallel Computing, Semester 2 2026

Assessment: Lab 2, Message Passing Interface (Week 8)

Team: Erwyna Soo Wen Xin (36555789, esoo0013@student.monash.edu) and Taabish Farooq Bhat (35473932, ttaa0006@student.monash.edu)

This part holds every prompt Erwyna typed into Claude for Lab 2, exported from the Claude Code session logs on 16 September 2026. The prompts are in the order they were sent, with the time in Malaysia time (UTC+8), and are copied exactly as typed, including typing and voice typing mistakes. Where a prompt contained a very long block of pasted course material, the middle of the pasted block is left out and marked. Answers picked in reply to Claude's multiple choice questions are included. Automatic system notifications and "request interrupted" markers are not, and neither are Claude's replies. The Gemini Pro prompt record is Part 2 of this file.

Session 1, 8 and 9 September 2026: reviewing Task 1 and Task 2, the slides, the Task 3 measurements, the CAAS run and packaging the submission

1. Prompt, Tuesday 8 September 2026, 3:20 PM

```
@"/Users/erwynasoo/Downloads/FIT3143-Topic-8-2024 (1).pps"x" @"/Users/erwynasoo/Desktop/FIT3143-Topic-8-2024.pdf"
@"/Users/erwynasoo/Desktop/Lab #2 Rubric.pdf" @"/Users/erwynasoo/Desktop/Lab #2 - Assessment Specification.pdf"
For FIT3143 look at what taabish has already completed in /Users/erwynasoo/FIT3143/Week8_Lab
```

```
* Would everyone welcome to 53 143 parallel computing.
0:03
* Uh, this is the extra class for week seven, and this extra class will cover some,
0:08
* uh, practical coding problems that would hopefully help you all in your lab.
0:14
* Week eight. All right. So I will turn off my camera.
0:19
* And if you have any questions, you can post it on the chat.
0:24
* Or you can unmute the microphone and ask your questions. Uh, during this extra class session can thank you
know.
0:27
* Um, I'm going to share my screen. Okay.
0:35
* So, um, there are several C files, and, uh, I will I
```

```
[... 140,116 characters of pasted course material left out here. The full prompt was 141,416 characters long
...]
```

```
h, for the applied sessions or tutorials, but that's about all we want to do.
53:31
* Uh, next section is going to talk about neural processing units.
53:38
* And. Much the same will apply. The intent here is to give you a survival guide.
53:44
* So we will resume with that shortly.
```

here are all the videos for W8 and content

I am in charge of task 3 and 4

create the slides in Canva accordingly

2. Prompt, Tuesday 8 September 2026, 4:27 PM

make me the slides NOW

3. Prompt, Tuesday 8 September 2026, 5:01 PM

<https://www.canva.com/d/Pt3SKVN8aq0jwjT>

WE like this design pls hone this with a script 7 minutes with lee way because we need to breathe

4. Prompt, Tuesday 8 September 2026, 10:36 PM

Could you just complete that vicious task so everything's done and dusted, um, and including the appendix? If it's needed, please do add it to my slides. And as for the graphs, I see some slides have, like, empty sections where they don't say it's graph one till nine, and it just says there, um, add diagram slots, stride assignment across ranks. Like, I don't know what to screenshot with that. So do help me on that. I only know, like, one, two, three, four, five, six, seven, Yeah. So help me on that. Yeah. finished the whole slides for me and make

sure that the, uh, everything is submission ready in my VSC code.

5. Prompt, Tuesday 8 September 2026, 10:53 PM

Ok and of course check the deck now I have updated it with all of the diagrams and graphs

Also look at my desktop and move the docs to vault accordingly

6. Prompt, Tuesday 8 September 2026, 11:19 PM

do clear the rest of the lab stuff from my desktop. I still see, like, the lab instructions there to my vault. And, um, could you just... is it possible to actually just, like, do tab issues part on your own and have me just, like, run it? But, yeah, it's not possible. When he does it, will he be able to submit on the GitHub his, uh, his findings, and will it be submittable in a way that the marker can read? As for everything else, could you also go to Canva and directly edit and insert our script into each slide, please? directly edited. Um, I made it so that anyone can edit the slide, including you. So here's the edit link.
<https://canva.link/r2mw03o8q0bmo5f>

7. Prompt, Wednesday 9 September 2026, 12:18 AM

I am connected to monash vpn:

```
erwynasoo@Erwynas-MacBook-Pro FIT3143 % sbatch run_all.job
zsh: command not found: sbatch
erwynasoo@Erwynas-MacBook-Pro FIT3143 % pip /Users/erwynasoo/FIT3143/Week8_Lab/caas/run_all.job
ERROR: unknown command "/Users/erwynasoo/FIT3143/Week8_Lab/caas/run_all.job"
erwynasoo@Erwynas-MacBook-Pro FIT3143 % npm run /Users/erwynasoo/FIT3143/Week8_Lab/caas/run_all.job
npm error code ENOENT
npm error syscall open
npm error path /Users/erwynasoo/FIT3143/package.json
npm error errno -2
npm error enoent Could not read package.json: Error: ENOENT: no such file or directory, open
'/Users/erwynasoo/FIT3143/package.json'
npm error enoent This is related to npm not being able to find a file.
npm error enoent
npm error A complete log of this run can be found in:
/Users/erwynasoo/.npm/_logs/2026-09-08T16_17_54_033Z-debug-0.log
erwynasoo@Erwynas-MacBook-Pro FIT3143 % pip8 /Users/erwynasoo/FIT3143/Week8_Lab/caas/run_all.job
zsh: command not found: pip8
erwynasoo@Erwynas-MacBook-Pro FIT3143 %
```

do i just run thsi and produce output?

8. Prompt, Wednesday 9 September 2026, 12:20 AM

give me the runnable commands my monash id is esoo0013 thats my email start my student ID u know it

9. Prompt, Wednesday 9 September 2026, 12:21 AM

how to find hostname

10. Prompt, Wednesday 9 September 2026, 12:24 AM

esoo0013@student.monash.edu>

is thsi my hostname my email

11. Prompt, Wednesday 9 September 2026, 12:26 AM

FIT3143 staff and students are registered on CAAS; and your user account is based on your Monash authcate credentials.

Your Monash authcate username is the word before the @ in your email address, e.g.:

abcd0001@student.monash.edu

The Remote Host address (using terminal):

ssh student-caas-headnode.rep.monash.edu (with IP 172.23.169.56)

When prompted for a username:

- * Use your Monash authcate username
- * Use your MyMonash password.

Reminder: You need to connect to Monash VPN before carrying out the step above. Refer to

<https://www.monash.edu/esolutions/network/vpn> for more information about connecting to Monash VPN. On successful SSH access into CAAS

Monash eSolutions - HPC-as-a-Service

Based on Red Hat Enterprise Linux 8

This system is hosted on the Monash Private Cloud. Access is restricted to Monash University authorised user

s. The use of this system is subject to \ Monash University's IT Acceptable Use Policy and is subject to regular monitoring. Unauthorised or improper

use of this system may result in administrative disciplinary action and/or civil and criminal penalties. Use of this system constitutes consent to security monitoring.

Use the following commands to adjust your environment:

```
'module avail'          - show available modules
'module add <module>'    - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login
-----
```

You will see a "Monash eSolutions - HPC-as-a-Service" message.

Below the message, you should see a command prompt, e.g.,
[yourAuthcate@student-caas-headnode ~]\$]
awaiting your textual commands.

14

12. Prompt, Wednesday 9 September 2026, 12:26 AM

FIT3143 staff and students are registered on CAAS; and your user account is based on your Monash authcate credentials. Your Monash authcate username is the word before the @ in your email address, e.g.:
abcd0001@student.monash.edu The Remote Host address (using terminal): ssh student-caas-headnode.rep.monash.edu (with IP 172.23.169.56) When prompted for a username:

- * Use your Monash authcate username
- * Use your MyMonash password.

Reminder: You need to connect to Monash VPN before carrying out the step above. Refer to <https://www.monash.edu/esolutions/network/vpn> for more information about connecting to Monash VPN. On successful SSH access into CAAS

Monash eSolutions - HPC-as-a-Service

Based on Red Hat Enterprise Linux 8

This system is hosted on the Monash Private Cloud. Access is restricted to Monash University authorised user s. The use of this system is subject to Monash University's IT Acceptable Use Policy and is subject to regular monitoring. Unauthorised or improper use of this system may result in administrative disciplinary action and/or civil and criminal penalties. Use of this system constitutes consent to security monitoring.

Use the following commands to adjust your environment:

```
'module avail' - show available modules
'module add <module>' - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login
```

You will see a "Monash eSolutions - HPC-as-a-Service" message.

Below the message, you should see a command prompt, e.g., [yourAuthcate@student-caas-headnode ~]\$] awaiting your textual commands.

14

my monash psw is student

13. Prompt, Wednesday 9 September 2026, 12:28 AM

```
ssh esoo0013@student-caas-headnode.rep.monash.edu
```

```
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
```

```
0013@student-caas-headnode.rep.monash.edu
```

The authenticity of host 'student-caas-headnode.rep.monash.edu (172.23.168.188)' can't be established.

ED25519 key fingerprint is SHA256:puesnNuuB2S+n13KeWdp2NEOXwkXyJWSVoJkzZ7iPOA.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

Warning: Permanently added 'student-caas-headnode.rep.monash.edu' (ED25519) to the list of known hosts.

```
esoo0013@student-caas-headnode.rep.monash.edu's password:
```

```
cd ~/Week8_Lab/caas
```

```
chmod +x run_all.job
```

```
sbatch run_all.job
```

Permission denied, please try again.

```
esoo0013@student-caas-headnode.rep.monash.edu's password:
```

sPermission denied, please try again.

```
esoo0013@student-caas-headnode.rep.monash.edu's password:
```

```
esoo0013@student-caas-headnode.rep.monash.edu: Permission denied
```

(publickey,gssapi-keyex,gssapi-with-mic,password).

```
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % Renaesoo
```

```
2005
```

```
zsh: command not found: Renaesoo2005
```

```
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo0013@student-caas-headnode.rep.monash.edu
esoo0013@student-caas-headnode.rep.monash.edu's password:
Welcome to Bright release          9.2
```

Based on Red Hat Enterprise Linux 8
ID: #389175

Use the following commands to adjust your environment:

```
'module avail'          - show available modules
'module add <module>'    - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login
```

```
-----
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last failed login: Wed Sep  9 02:28:08 AEST 2026 from 10.44.128.131 on ssh:notty
There were 3 failed login attempts since the last successful login.
Creating ECDSA key for ssh
[esoo0013@student-caas-headnode ~]$ cd ~/Week8_Lab/caas
-bash: cd: /srv/home/esoo0013/Week8_Lab/caas: No such file or directory
[esoo0013@student-caas-headnode ~]$ chmod +x run_all.job
chmod: cannot access 'run_all.job': No such file or directory
[esoo0013@student-caas-headnode ~]$ sbatch run_all.job
sbatch: error: Unable to open file run_all.job
[esoo0013@student-caas-headnode ~]$ squeue -u esoo0013
      JOBID PARTITION    NAME    USER ST     TIME  NODES NODELIST(REASON)
[esoo0013@student-caas-headnode ~]$ cat caas_results.txt
cat: caas_results.txt: No such file or directory
[esoo0013@student-caas-headnode ~]$
```

14. Prompt, Wednesday 9 September 2026, 2:55 PM

```
squeue -u esoo0013
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % squeue -
u esoo0013
zsh: command not found: squeue
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas
_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % squeue -
u esoo0013
zsh: command not found: squeue
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cd ~/Week8_Lab/caas
chmod +x run_all.job
sbatch run_all.job
cd: no such file or directory: /Users/erwynasoo/Week8_Lab/caas
chmod: run_all.job: No such file or directory
zsh: command not found: sbatch
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault %
```

15. Prompt, Wednesday 9 September 2026, 3:04 PM

```
Alright scp -r /Users/erwynasoo/FIT3143/Week8_Lab esoo0013@student-caas-headnode.rep.monash.edu:~/
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % scp -r /
Users/erwynasoo/FIT3143/Week8_Lab esoo0013@student-caas
-headnode.rep.monash.edu:~/
esoo0013@student-caas-headnode.rep.monash.edu's password:
.Rhistry          100%    0      0.0KB/s   00:00
task_2_instr.c    100% 7901    45.8KB/s   00:00
primes_omp.txt    100%  64MB    1.1MB/s    00:57
primes_task1_instr.t 100%  64MB 785.1KB/s   01:23
task_2.c          100% 6706    35.8KB/s   00:00
make_task3_doc.py 100%  18KB   63.7KB/s   00:00
launch_overhead.txt 100%    9      0.0KB/s    00:00
results_by_procs.csv 100%  753     4.0KB/s    00:00
primes_pthread.txt 100%  64MB    1.2MB/s    00:54
run_benchmarks.sh 100%  11KB   51.8KB/s    00:00
task_1_instr.c    100% 8794    48.2KB/s    00:00
serial_instr.c    100% 3470    18.7KB/s    00:00
Task3_Performance_Ev 100%  16KB   80.7KB/s    00:00
task_1.c          100% 5395    28.0KB/s    00:00
primes_serial_instr. 100%  35MB 633.0KB/s    00:56
primes_serial.txt 100% 2635KB   1.1MB/s    00:02
Presentation_Script. 100% 9746    50.6KB/s    00:00
run_all.job       100% 4144    19.3KB/s    00:00
hybrid.job        100%  927     4.9KB/s    00:00
README.md         100% 3486    20.6KB/s    00:00
serial.job        100%  764     4.0KB/s    00:00
```

```

.gitkeep                100%    0      0.0KB/s   00:00
mpi.job                 100% 1192      7.0KB/s   00:00
run_phases.py           100% 2931     13.5KB/s   00:00
README.md               100% 5091     29.1KB/s   00:00
Presentation_Slides_    100% 3519KB    1.6MB/s   00:02
.gitignore              100%  164      1.0KB/s   00:00
make_diagrams.py        100% 9563     17.9KB/s   00:00
make_graphs.py          100%  19KB     36.5KB/s   00:00
AI_Declaration.md       100% 2717     16.0KB/s   00:00
measure_launch.py       100% 1692      9.5KB/s   00:00
phases_serial.csv       100%  193      1.2KB/s   00:00
results_hybrid_t.csv    100%  175      0.8KB/s   00:00
primes_task1.txt        100% 2635KB    867.6KB/s  00:03
diagram1_stride_part    100% 239KB    556.6KB/s  00:00
graph3_speedup_vs_wi    100% 267KB    657.7KB/s  00:00
graph2_speedup_vs_n.    100% 342KB    847.6KB/s  00:00
diagram3_hybrid_layo    100% 234KB    600.1KB/s  00:00
graph5_matched_width    100% 405KB    824.5KB/s  00:00
graph6_amdahl_task1.    100% 330KB    280.1KB/s  00:01
diagram2_gatherv_off    100% 297KB     79.4KB/s  00:03
graph1_runtime_vs_n.    100% 289KB    318.1KB/s  00:00
graph8_phase_breakdo    100% 221KB    347.1KB/s  00:00
graph7_amdahl_task2.    100% 333KB    383.7KB/s  00:00
graph4_hybrid_vs_mpi    100% 218KB    599.2KB/s  00:00
graph9_rank_balance.    100% 156KB    455.4KB/s  00:00
results_hybrid_total    100%  564      3.3KB/s   00:00
mpi                     100%  34KB     95.1KB/s   00:00
hybrid_instr            100%  35KB     85.0KB/s   00:00
pthread                 100%  50KB    123.0KB/s  00:00
omp                     100%  34KB     67.6KB/s   00:00
ref.txt                 100% 15MB    440.8KB/s  00:35
mpi_instr               100%  34KB     86.5KB/s   00:00
serial_instr            100%  33KB     64.1KB/s   00:00
serial                  100%  33KB     63.2KB/s   00:00
hybrid                  100%  34KB     96.6KB/s   00:00
rank_balance.csv        100%  758      4.1KB/s   00:00
amdahl_summary.csv      100% 1775      5.8KB/s   00:00
primes_task2.txt        100% 2635KB    1.6MB/s   00:01
primes_task2_instr.t    100%  64MB     1.5MB/s   00:43
task_2.c.orig-backup    100% 5445     30.0KB/s   00:00
results_by_n.csv        100% 2002     10.8KB/s   00:00
phases_task2.csv        100% 1569      8.4KB/s   00:00
bench.log               100% 6588     33.5KB/s   00:00
phases_task1.csv        100% 1968     11.0KB/s   00:00
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault %

```

what happens now I dont see anything submitted or changed in vsc

16. Prompt, Wednesday 9 September 2026, 3:05 PM

```

scp -r /Users/erwynasoo/FIT3143/Week8_Lab esoo0013@student-caas-headnode.rep.monash.edu:~/
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % scp -r /
Users/erwynasoo/FIT3143/Week8_Lab esoo0013@student-caas
-headnode.rep.monash.edu:~/
esoo0013@student-caas-headnode.rep.monash.edu's password:
.Rhistory                100%    0      0.0KB/s   00:00
task_2_instr.c           100% 7901     45.8KB/s   00:00
primes_omp.txt           100%  64MB     1.1MB/s   00:57
primes_task1_instr.t     100%  64MB    785.1KB/s  01:23
task_2.c                 100% 6706     35.8KB/s   00:00
make_task3_doc.py        100%  18KB     63.7KB/s   00:00
launch_overhead.txt      100%    9      0.0KB/s   00:00
results_by_procs.csv     100%  753      4.0KB/s   00:00
primes_pthread.txt       100%  64MB     1.2MB/s   00:54
run_benchmarks.sh        100%  11KB     51.8KB/s   00:00
task_1_instr.c           100% 8794     48.2KB/s   00:00
serial_instr.c           100% 3470     18.7KB/s   00:00
Task3_Performance_Ev     100%  16KB     80.7KB/s   00:00
task_1.c                 100% 5395     28.0KB/s   00:00
primes_serial_instr.     100%  35MB    633.0KB/s  00:56
primes_serial.txt        100% 2635KB    1.1MB/s   00:02
Presentation_Script.     100% 9746     50.6KB/s   00:00
run_all.job              100% 4144     19.3KB/s   00:00
hybrid.job               100%  927      4.9KB/s   00:00
README.md                100% 3486     20.6KB/s   00:00
serial.job               100%  764      4.0KB/s   00:00
.gitkeep                 100%    0      0.0KB/s   00:00
mpi.job                  100% 1192      7.0KB/s   00:00

```

```

run_phases.py          100% 2931    13.5KB/s   00:00
README.md              100% 5091    29.1KB/s   00:00
Presentation_Slides_   100% 3519KB   1.6MB/s   00:02
.gitignore             100% 164      1.0KB/s   00:00
make_diagrams.py       100% 9563    17.9KB/s   00:00
make_graphs.py         100% 19KB     36.5KB/s   00:00
AI_Declaration.md      100% 2717    16.0KB/s   00:00
measure_launch.py      100% 1692     9.5KB/s   00:00
phases_serial.csv      100% 193      1.2KB/s   00:00
results_hybrid_t.csv   100% 175      0.8KB/s   00:00
primes_task1.txt       100% 2635KB  867.6KB/s  00:03
diagram1_stride_part   100% 239KB   556.6KB/s  00:00
graph3_speedup_vs_wi   100% 267KB   657.7KB/s  00:00
graph2_speedup_vs_n.   100% 342KB   847.6KB/s  00:00
diagram3_hybrid_layo   100% 234KB   600.1KB/s  00:00
graph5_matched_width   100% 405KB   824.5KB/s  00:00
graph6_amdahl_task1.   100% 330KB   280.1KB/s  00:01
diagram2_gatherv_off   100% 297KB    79.4KB/s  00:03
graph1_runtime_vs_n.   100% 289KB   318.1KB/s  00:00
graph8_phase_breakdo   100% 221KB   347.1KB/s  00:00
graph7_amdahl_task2.   100% 333KB   383.7KB/s  00:00
graph4_hybrid_vs_mpi   100% 218KB   599.2KB/s  00:00
graph9_rank_balance.   100% 156KB   455.4KB/s  00:00
results_hybrid_total   100% 564      3.3KB/s   00:00
mpi                    100% 34KB    95.1KB/s   00:00
hybrid_instr           100% 35KB    85.0KB/s   00:00
pthread               100% 50KB   123.0KB/s  00:00
omp                    100% 34KB    67.6KB/s   00:00
ref.txt               100% 15MB   440.8KB/s  00:35
mpi_instr              100% 34KB    86.5KB/s   00:00
serial_instr           100% 33KB    64.1KB/s   00:00
serial                 100% 33KB    63.2KB/s   00:00
hybrid                 100% 34KB    96.6KB/s   00:00
rank_balance.csv       100% 758      4.1KB/s   00:00
amdahl_summary.csv     100% 1775     5.8KB/s   00:00
primes_task2.txt       100% 2635KB   1.6MB/s   00:01
primes_task2_instr.t   100% 64MB    1.5MB/s   00:43
task_2.c.orig-backup   100% 5445    30.0KB/s   00:00
results_by_n.csv       100% 2002    10.8KB/s   00:00
phases_task2.csv       100% 1569     8.4KB/s   00:00
bench.log              100% 6588    33.5KB/s   00:00
phases_task1.csv       100% 1968    11.0KB/s   00:00
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
0013@student-caas-headnode.rep.monash.edu
esoo0013@student-caas-headnode.rep.monash.edu's password:
Welcome to Bright release          9.2

```

Based on Red Hat Enterprise Linux 8
ID: #389175

Use the following commands to adjust your environment:

```

'module avail'          - show available modules
'module add <module>'   - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login

```

```

-----
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Wed Sep  9 16:54:57 2026 from 10.44.128.159
[esoo0013@student-caas-headnode ~]$ cd ~/Week8_Lab/caas
[esoo0013@student-caas-headnode caas]$ chmod +x run_all.job
[esoo0013@student-caas-headnode caas]$ sbatch run_all.job
Submitted batch job 39354
[esoo0013@student-caas-headnode caas]$ squeue -u esoo0013
      JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
[esoo0013@student-caas-headnode caas]$

```

17. Prompt, Wednesday 9 September 2026, 3:06 PM

```

scp -r /Users/erwynasoo/FIT3143/Week8_Lab esoo0013@student-caas-headnode.rep.monash.edu:~/
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % scp -r /
Users/erwynasoo/FIT3143/Week8_Lab esoo0013@student-caas
-headnode.rep.monash.edu:~/
esoo0013@student-caas-headnode.rep.monash.edu's password:
.Rhistory          100%  0      0.0KB/s   00:00
task_2_instr.c     100% 7901    45.8KB/s   00:00
primes_omp.txt     100% 64MB    1.1MB/s   00:57

```



```

primes_task1_instr.t 100% 64MB 785.1KB/s 01:23
task_2.c 100% 6706 35.8KB/s 00:00
make_task3_doc.py 100% 18KB 63.7KB/s 00:00
launch_overhead.txt 100% 9 0.0KB/s 00:00
results_by_procs.csv 100% 753 4.0KB/s 00:00
primes_pthread.txt 100% 64MB 1.2MB/s 00:54
run_benchmarks.sh 100% 11KB 51.8KB/s 00:00
task_1_instr.c 100% 8794 48.2KB/s 00:00
serial_instr.c 100% 3470 18.7KB/s 00:00
Task3_Performance_Ev 100% 16KB 80.7KB/s 00:00
task_1.c 100% 5395 28.0KB/s 00:00
primes_serial_instr. 100% 35MB 633.0KB/s 00:56
primes_serial.txt 100% 2635KB 1.1MB/s 00:02
Presentation_Script. 100% 9746 50.6KB/s 00:00
run_all.job 100% 4144 19.3KB/s 00:00
hybrid.job 100% 927 4.9KB/s 00:00
README.md 100% 3486 20.6KB/s 00:00
serial.job 100% 764 4.0KB/s 00:00
.gitkeep 100% 0 0.0KB/s 00:00
mpi.job 100% 1192 7.0KB/s 00:00
run_phases.py 100% 2931 13.5KB/s 00:00
README.md 100% 5091 29.1KB/s 00:00
Presentation_Slides_ 100% 3519KB 1.6MB/s 00:02
.gitignore 100% 164 1.0KB/s 00:00
make_diagrams.py 100% 9563 17.9KB/s 00:00
make_graphs.py 100% 19KB 36.5KB/s 00:00
AI_Declaration.md 100% 2717 16.0KB/s 00:00
measure_launch.py 100% 1692 9.5KB/s 00:00
phases_serial.csv 100% 193 1.2KB/s 00:00
results_hybrid_t.csv 100% 175 0.8KB/s 00:00
primes_task1.txt 100% 2635KB 867.6KB/s 00:03
diagram1_stride_part 100% 239KB 556.6KB/s 00:00
graph3_speedup_vs_wi 100% 267KB 657.7KB/s 00:00
graph2_speedup_vs_n. 100% 342KB 847.6KB/s 00:00
diagram3_hybrid_layo 100% 234KB 600.1KB/s 00:00
graph5_matched_width 100% 405KB 824.5KB/s 00:00
graph6_amdahl_task1. 100% 330KB 280.1KB/s 00:01
diagram2_gatherv_off 100% 297KB 79.4KB/s 00:03
graph1_runtime_vs_n. 100% 289KB 318.1KB/s 00:00
graph8_phase_breakdo 100% 221KB 347.1KB/s 00:00
graph7_amdahl_task2. 100% 333KB 383.7KB/s 00:00
graph4_hybrid_vs_mpi 100% 218KB 599.2KB/s 00:00
graph9_rank_balance. 100% 156KB 455.4KB/s 00:00
results_hybrid_total 100% 564 3.3KB/s 00:00
mpi 100% 34KB 95.1KB/s 00:00
hybrid_instr 100% 35KB 85.0KB/s 00:00
pthread 100% 50KB 123.0KB/s 00:00
omp 100% 34KB 67.6KB/s 00:00
ref.txt 100% 15MB 440.8KB/s 00:35
mpi_instr 100% 34KB 86.5KB/s 00:00
serial_instr 100% 33KB 64.1KB/s 00:00
serial 100% 33KB 63.2KB/s 00:00
hybrid 100% 34KB 96.6KB/s 00:00
rank_balance.csv 100% 758 4.1KB/s 00:00
amdahl_summary.csv 100% 1775 5.8KB/s 00:00
primes_task2.txt 100% 2635KB 1.6MB/s 00:01
primes_task2_instr.t 100% 64MB 1.5MB/s 00:43
task_2.c.orig-backup 100% 5445 30.0KB/s 00:00
results_by_n.csv 100% 2002 10.8KB/s 00:00
phases_task2.csv 100% 1569 8.4KB/s 00:00
bench.log 100% 6588 33.5KB/s 00:00
phases_task1.csv 100% 1968 11.0KB/s 00:00
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
0013@student-caas-headnode.rep.monash.edu
esoo0013@student-caas-headnode.rep.monash.edu's password:
Welcome to Bright release 9.2

```

Based on Red Hat Enterprise Linux 8
ID: #389175

Use the following commands to adjust your environment:

```

'module avail'          - show available modules
'module add <module>'   - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login

```

```

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Wed Sep  9 16:54:57 2026 from 10.44.128.159
[esoo0013@student-caas-headnode ~]$ cd ~/Week8_Lab/caas
[esoo0013@student-caas-headnode caas]$ chmod +x run_all.job
[esoo0013@student-caas-headnode caas]$ sbatch run_all.job
Submitted batch job 39354
[esoo0013@student-caas-headnode caas]$ squeue -u esoo0013
      JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
[esoo0013@student-caas-headnode caas]$ cat caas_results.txt
cat: caas_results.txt: No such file or directory
[esoo0013@student-caas-headnode caas]$ cat caas_results.txt
cat: caas_results.txt: No such file or directory
[esoo0013@student-caas-headnode caas]$

```

18. Prompt, Wednesday 9 September 2026, 3:07 PM

```

cat caas_results.txt
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas
_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ls -la l
ab2_caas*.out
zsh: no matches found: lab2_caas*.out
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat lab2
_caas.39354.out
cat: lab2_caas.39354.out: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault %

```

19. Prompt, Wednesday 9 September 2026, 3:09 PM

```

cat caas_results.txt
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas
_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ls -la l
ab2_caas*.out
zsh: no matches found: lab2_caas*.out
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat lab2
_caas.39354.out
cat: lab2_caas.39354.out: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
0013@student-caas-headnode.rep.monash.edu
esoo0013@student-caas-headnode.rep.monash.edu's password:
Welcome to Bright release          9.2

```

Based on Red Hat Enterprise Linux 8
ID: #389175

Use the following commands to adjust your environment:

```

'module avail'           - show available modules
'module add <module>'    - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login

```

```

-----
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Wed Sep  9 17:05:25 2026 from 10.44.128.159
[esoo0013@student-caas-headnode ~]$ cd ~/Week8_Lab/caas
[esoo0013@student-caas-headnode caas]$ ls -la
total 44
drwxr-xr-x 3 esoo0013 student 4096 Sep  9 17:05 .
drwxr-xr-x 5 esoo0013 student 4096 Sep  9 02:30 ..
-rwxr-xr-x 1 esoo0013 student  927 Sep  9 17:01 hybrid.job
-rw-r--r-- 1 esoo0013 student  123 Sep  9 16:55 lab2_caas.39353.out
-rw-r--r-- 1 esoo0013 student  123 Sep  9 17:05 lab2_caas.39354.out
-rwxr-xr-x 1 esoo0013 student 1192 Sep  9 17:01 mpi.job
-rw-r--r-- 1 esoo0013 student 3486 Sep  9 17:01 README.md
drwxr-xr-x 2 esoo0013 student 4096 Sep  9 02:30 results
-rwxr-xr-x 1 esoo0013 student 4144 Sep  9 17:01 run_all.job
-rwxr-xr-x 1 esoo0013 student  764 Sep  9 17:01 serial.job
[esoo0013@student-caas-headnode caas]$

```

20. Prompt, Wednesday 9 September 2026, 3:09 PM

```

cat caas_results.txt
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas
_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ls -la l
ab2_caas*.out
zsh: no matches found: lab2_caas*.out
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat lab2
_caas.39354.out
cat: lab2_caas.39354.out: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
0013@student-caas-headnode.rep.monash.edu
esoo0013@student-caas-headnode.rep.monash.edu's password:
Welcome to Bright release          9.2

```

Based on Red Hat Enterprise Linux 8
ID: #389175

Use the following commands to adjust your environment:

```

'module avail'          - show available modules
'module add <module>'   - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login

```

```

-----
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Wed Sep  9 17:05:25 2026 from 10.44.128.159
[esoo0013@student-caas-headnode ~]$ cd ~/Week8_Lab/caas
[esoo0013@student-caas-headnode caas]$ ls -la
total 44
drwxr-xr-x 3 esoo0013 student 4096 Sep  9 17:05 .
drwxr-xr-x 5 esoo0013 student 4096 Sep  9 02:30 ..
-rwxr-xr-x 1 esoo0013 student  927 Sep  9 17:01 hybrid.job
-rw-r--r-- 1 esoo0013 student  123 Sep  9 16:55 lab2_caas.39353.out
-rw-r--r-- 1 esoo0013 student  123 Sep  9 17:05 lab2_caas.39354.out
-rwxr-xr-x 1 esoo0013 student 1192 Sep  9 17:01 mpi.job
-rw-r--r-- 1 esoo0013 student 3486 Sep  9 17:01 README.md
drwxr-xr-x 2 esoo0013 student 4096 Sep  9 02:30 results
-rwxr-xr-x 1 esoo0013 student 4144 Sep  9 17:01 run_all.job
-rwxr-xr-x 1 esoo0013 student  764 Sep  9 17:01 serial.job
[esoo0013@student-caas-headnode caas]$ cat lab2_caas.39354.out
Building...
ccl: fatal error: ../../Lab1_W4/task1.c: No such file or directory
compilation terminated.
serial build FAILED
[esoo0013@student-caas-headnode caas]$

```

21. Prompt, Wednesday 9 September 2026, 3:15 PM

```

cat caas_results.txt
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas
_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ls -la l
ab2_caas*.out
zsh: no matches found: lab2_caas*.out
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat lab2
_caas.39354.out
cat: lab2_caas.39354.out: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
0013@student-caas-headnode.rep.monash.edu
esoo0013@student-caas-headnode.rep.monash.edu's password:
Welcome to Bright release          9.2

```

Based on Red Hat Enterprise Linux 8
ID: #389175

Use the following commands to adjust your environment:

```

'module avail'          - show available modules
'module add <module>'   - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login

```

```

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Wed Sep  9 17:05:25 2026 from 10.44.128.159
[esoo0013@student-caas-headnode ~]$ cd ~/Week8_Lab/caas
[esoo0013@student-caas-headnode caas]$ ls -la
total 44
drwxr-xr-x 3 esoo0013 student 4096 Sep  9 17:05 .
drwxr-xr-x 5 esoo0013 student 4096 Sep  9 02:30 ..
-rwxr-xr-x 1 esoo0013 student  927 Sep  9 17:01 hybrid.job
-rw-r--r-- 1 esoo0013 student  123 Sep  9 16:55 lab2_caas.39353.out
-rw-r--r-- 1 esoo0013 student  123 Sep  9 17:05 lab2_caas.39354.out
-rwxr-xr-x 1 esoo0013 student 1192 Sep  9 17:01 mpi.job
-rw-r--r-- 1 esoo0013 student 3486 Sep  9 17:01 README.md
drwxr-xr-x 2 esoo0013 student 4096 Sep  9 02:30 results
-rwxr-xr-x 1 esoo0013 student 4144 Sep  9 17:01 run_all.job
-rwxr-xr-x 1 esoo0013 student  764 Sep  9 17:01 serial.job
[esoo0013@student-caas-headnode caas]$ cat lab2_caas.39354.out
Building...
ccl: fatal error: ../../Lab1_W4/task1.c: No such file or directory
compilation terminated.
serial build FAILED
[esoo0013@student-caas-headnode caas]$ ssh esoo0013@student-caas-headnode.rep.monash.edu "mkdir -p ~/Lab1_W4"
Warning: Permanently added 'student-caas-headnode.rep.monash.edu,172.23.168.188' (ECDSA) to the list of known
hosts.

[esoo0013@student-caas-headnode caas]$ cd ~/Week8_Lab/caas
[esoo0013@student-caas-headnode caas]$ sbatch run_all.job
Submitted batch job 39355
[esoo0013@student-caas-headnode caas]$ squeue -u esoo0013
          JOBID PARTITION    NAME    USER  ST       TIME  NODES NODELIST(REASON)
[esoo0013@student-caas-headnode caas]$

```

22. Prompt, Wednesday 9 September 2026, 3:26 PM

```

cat caas_results.txt
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas
_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ls -la l
ab2_caas*.out
zsh: no matches found: lab2_caas*.out
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat lab2
_caas.39354.out
cat: lab2_caas.39354.out: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
0013@student-caas-headnode.rep.monash.edu
esoo0013@student-caas-headnode.rep.monash.edu's password:
Welcome to Bright release          9.2

```

Based on Red Hat Enterprise Linux 8
ID: #389175

Use the following commands to adjust your environment:

```

'module avail'           - show available modules
'module add <module>'    - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login

```

```

-----
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Wed Sep  9 17:05:25 2026 from 10.44.128.159
[esoo0013@student-caas-headnode ~]$ cd ~/Week8_Lab/caas
[esoo0013@student-caas-headnode caas]$ ls -la
total 44
drwxr-xr-x 3 esoo0013 student 4096 Sep  9 17:05 .
drwxr-xr-x 5 esoo0013 student 4096 Sep  9 02:30 ..
-rwxr-xr-x 1 esoo0013 student  927 Sep  9 17:01 hybrid.job
-rw-r--r-- 1 esoo0013 student  123 Sep  9 16:55 lab2_caas.39353.out
-rw-r--r-- 1 esoo0013 student  123 Sep  9 17:05 lab2_caas.39354.out
-rwxr-xr-x 1 esoo0013 student 1192 Sep  9 17:01 mpi.job
-rw-r--r-- 1 esoo0013 student 3486 Sep  9 17:01 README.md
drwxr-xr-x 2 esoo0013 student 4096 Sep  9 02:30 results
-rwxr-xr-x 1 esoo0013 student 4144 Sep  9 17:01 run_all.job
-rwxr-xr-x 1 esoo0013 student  764 Sep  9 17:01 serial.job
[esoo0013@student-caas-headnode caas]$ cat lab2_caas.39354.out

```

```

Building...
ccl: fatal error: ../../Lab1_W4/task1.c: No such file or directory
compilation terminated.
serial build FAILED
[esoo0013@student-caas-headnode caas]$ ssh esoo0013@student-caas-headnode.rep.monash.edu "mkdir -p ~/Lab1_W4"
Warning: Permanently added 'student-caas-headnode.rep.monash.edu,172.23.168.188' (ECDSA) to the list of known
hosts.

[esoo0013@student-caas-headnode caas]$ cd ~/Week8_Lab/caas
[esoo0013@student-caas-headnode caas]$ sbatch run_all.job
Submitted batch job 39355
[esoo0013@student-caas-headnode caas]$ squeue -u esoo0013
          JOBID PARTITION    NAME    USER ST       TIME  NODES NODELIST(REASON)
[esoo0013@student-caas-headnode caas]$ scp /Users/erwynasoo/FIT3143/Lab1_W4/task1.c
esoo0013@student-caas-headnode.rep.monash.edu:~/Lab1_W4/
/Users/erwynasoo/FIT3143/Lab1_W4/task1.c: No such file or directory
[esoo0013@student-caas-headnode caas]$ sbatch run_all.job
Submitted batch job 39357
[esoo0013@student-caas-headnode caas]$ squeue -u esoo0013
          JOBID PARTITION    NAME    USER ST       TIME  NODES NODELIST(REASON)
          39357      defq fit3143_ esoo0013  R        0:00        2 student-caas-n[01-02]
[esoo0013@student-caas-headnode caas]$

```

23. Prompt, Wednesday 9 September 2026, 3:28 PM

```

cat caas_results.txt
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas
_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat caas_results.txt
cat: caas_results.txt: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ls -la l
ab2_caas.*.out
zsh: no matches found: lab2_caas.*.out
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cat lab2
_caas.39354.out
cat: lab2_caas.39354.out: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
0013@student-caas-headnode.rep.monash.edu
esoo0013@student-caas-headnode.rep.monash.edu's password:
Welcome to Bright release          9.2

```

Based on Red Hat Enterprise Linux 8
ID: #389175

Use the following commands to adjust your environment:

```

'module avail'          - show available modules
'module add <module>'   - adds a module to your environment for this session
'module initadd <module>' - configure module to be loaded at every login

```

```

-----
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Wed Sep  9 17:05:25 2026 from 10.44.128.159
[esoo0013@student-caas-headnode ~]$ cd ~/Week8_Lab/caas
[esoo0013@student-caas-headnode caas]$ ls -la
total 44
drwxr-xr-x 3 esoo0013 student 4096 Sep  9 17:05 .
drwxr-xr-x 5 esoo0013 student 4096 Sep  9 02:30 ..
-rwxr-xr-x 1 esoo0013 student  927 Sep  9 17:01 hybrid.job
-rw-r--r-- 1 esoo0013 student  123 Sep  9 16:55 lab2_caas.39353.out
-rw-r--r-- 1 esoo0013 student  123 Sep  9 17:05 lab2_caas.39354.out
-rwxr-xr-x 1 esoo0013 student 1192 Sep  9 17:01 mpi.job
-rw-r--r-- 1 esoo0013 student 3486 Sep  9 17:01 README.md
drwxr-xr-x 2 esoo0013 student 4096 Sep  9 02:30 results
-rwxr-xr-x 1 esoo0013 student 4144 Sep  9 17:01 run_all.job
-rwxr-xr-x 1 esoo0013 student  764 Sep  9 17:01 serial.job
[esoo0013@student-caas-headnode caas]$ cat lab2_caas.39354.out
Building...
ccl: fatal error: ../../Lab1_W4/task1.c: No such file or directory
compilation terminated.
serial build FAILED
[esoo0013@student-caas-headnode caas]$ ssh esoo0013@student-caas-headnode.rep.monash.edu "mkdir -p ~/Lab1_W4"
Warning: Permanently added 'student-caas-headnode.rep.monash.edu,172.23.168.188' (ECDSA) to the list of known
hosts.

[esoo0013@student-caas-headnode caas]$ cd ~/Week8_Lab/caas

```

```
[esoo0013@student-caas-headnode caas]$ sbatch run_all.job
Submitted batch job 39355
[esoo0013@student-caas-headnode caas]$ squeue -u esoo0013
      JOBID PARTITION    NAME    USER ST       TIME  NODES NODELIST(REASON)
[esoo0013@student-caas-headnode caas]$ scp /Users/erwynasoo/FIT3143/Lab1_W4/task1.c
esoo0013@student-caas-headnode.rep.monash.edu:~/Lab1_W4/
/Users/erwynasoo/FIT3143/Lab1_W4/task1.c: No such file or directory
[esoo0013@student-caas-headnode caas]$ sbatch run_all.job
Submitted batch job 39357
[esoo0013@student-caas-headnode caas]$ squeue -u esoo0013
      JOBID PARTITION    NAME    USER ST       TIME  NODES NODELIST(REASON)
    39357      defq fit3143_ esoo0013  R       0:00      2 student-caas-n[01-02]
[esoo0013@student-caas-headnode caas]$ cat lab2_caas.39357.out
Building...
ccl: fatal error: ../../Lab1_W4/task1.c: No such file or directory
compilation terminated.
serial build FAILED
[esoo0013@student-caas-headnode caas]$ squeue -u esoo0013
      JOBID PARTITION    NAME    USER ST       TIME  NODES NODELIST(REASON)
[esoo0013@student-caas-headnode caas]$ scp /Users/erwynasoo/FIT3143/Lab1_W4/task1.c
esoo0013@student-caas-headnode.rep.monash.edu:~/Lab1_W4/
/Users/erwynasoo/FIT3143/Lab1_W4/task1.c: No such file or directory
[esoo0013@student-caas-headnode caas]$ cd ~/Week8_Lab/caas
[esoo0013@student-caas-headnode caas]$ sbatch run_all.job
Submitted batch job 39360
[esoo0013@student-caas-headnode caas]$ squeue -u esoo0013
      JOBID PARTITION    NAME    USER ST       TIME  NODES NODELIST(REASON)
    39360      defq fit3143_ esoo0013  PD       0:00      2 (Priority)
[esoo0013@student-caas-headnode caas]$
```

why lab 4? it is lab 8

24. Prompt, Wednesday 9 September 2026, 3:38 PM

```
scancel 39360
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % scancel
39360
zsh: command not found: scancel
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cd ~/Week8_Lab/caas
sed -i 's|\\.\./\./Lab1_W4/task1\.c|../serial_instr.c|; s|primes_serial\.txt|primes_serial_instr.txt|'
run_all.job
cd: no such file or directory: /Users/erwynasoo/Week8_Lab/caas
sed: -I or -i may not be used with stdin
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % grep -n "caas_serial\\|primes_serial" run_all.job
grep: run_all.job: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % grep -n
"caas_serial\\|primes_serial" run_all.job
grep: run_all.job: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % sbatch run_all.job
squeue -u esoo0013
zsh: command not found: sbatch
zsh: command not found: squeue
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % squeue -u esoo0013
zsh: command not found: squeue
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault %
```

25. Prompt, Wednesday 9 September 2026, 3:40 PM

```
scancel 39360
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % scancel
39360
zsh: command not found: scancel
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cd ~/Week8_Lab/caas
sed -i 's|\\.\./\./Lab1_W4/task1\.c|../serial_instr.c|; s|primes_serial\.txt|primes_serial_instr.txt|'
run_all.job
cd: no such file or directory: /Users/erwynasoo/Week8_Lab/caas
sed: -I or -i may not be used with stdin
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % grep -n "caas_serial\\|primes_serial" run_all.job
grep: run_all.job: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % grep -n
"caas_serial\\|primes_serial" run_all.job
grep: run_all.job: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % sbatch run_all.job
squeue -u esoo0013
zsh: command not found: sbatch
zsh: command not found: squeue
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % squeue -u esoo0013
zsh: command not found: squeue
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
```

```

0013@student-caas-headnode.rep.monash.edu 'scancel 3936
0 2>/dev/null; cd ~/Week8_Lab/caas && sed -i "s|\\.\.\\.
\\.Lab1_W4/task1\\.c|../serial_instr.c|; s|primes_serial
\\.txt|primes_serial_instr.txt|" run_all.job && echo "--
- patched lines ---" && grep -n "caas_serial\\|primes_se
rial" run_all.job && echo "--- submitting ---" && sbatc
h run_all.job && sleep 20 && echo "--- queue after 20s
---" && queue -u esoo0013'
esoo0013@student-caas-headnode.rep.monash.edu's password:
--- patched lines ---
45:gcc -O2 ../serial_instr.c -o caas_serial -lm || { echo "serial build FAILED"; exit 1; }
62: ./caas_serial $N
84: tail -n +2 primes_serial_instr.txt > ref.txt 2>/dev/null
--- submitting ---
Submitted batch job 39361

```

26. Prompt, Wednesday 9 September 2026, 3:41 PM

```

scancel 39360
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % scancel
39360
zsh: command not found: scancel
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cd ~/Week8_Lab/caas
sed -i 's|\\.\.\\.Lab1_W4/task1\\.c|../serial_instr.c|; s|primes_serial\\.txt|primes_serial_instr.txt|'
run_all.job
cd: no such file or directory: /Users/erwynasoo/Week8_Lab/caas
sed: -I or -i may not be used with stdin
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % grep -n "caas_serial\\|primes_serial" run_all.job
grep: run_all.job: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % grep -n
"caas_serial\\|primes_serial" run_all.job
grep: run_all.job: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % sbatch run_all.job
queue -u esoo0013
zsh: command not found: sbatch
zsh: command not found: queue
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % queue -u esoo0013
zsh: command not found: queue
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
0013@student-caas-headnode.rep.monash.edu 'scancel 3936
0 2>/dev/null; cd ~/Week8_Lab/caas && sed -i "s|\\.\.\\.
\\.Lab1_W4/task1\\.c|../serial_instr.c|; s|primes_serial
\\.txt|primes_serial_instr.txt|" run_all.job && echo "--
- patched lines ---" && grep -n "caas_serial\\|primes_se
rial" run_all.job && echo "--- submitting ---" && sbatc
h run_all.job && sleep 20 && echo "--- queue after 20s
---" && queue -u esoo0013'
esoo0013@student-caas-headnode.rep.monash.edu's password:
--- patched lines ---
45:gcc -O2 ../serial_instr.c -o caas_serial -lm || { echo "serial build FAILED"; exit 1; }
62: ./caas_serial $N
84: tail -n +2 primes_serial_instr.txt > ref.txt 2>/dev/null
--- submitting ---
Submitted batch job 39361
--- queue after 20s ---

```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	ODELIST(Reason)
39361	defq	fit3143_ esoo0013	R		0:19	2	student-caas-n[01-02]

```

erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo0013@student-caas-headnode.rep.monash.edu 'echo "---
queue ---"; queue -u esoo0013; echo "--- latest log ---"; cd ~/Week8_Lab/caas && ls -t lab2_caas.*.out | head
-1 | xargs tail -30'
esoo0013@student-caas-headnode.rep.monash.edu's password:
--- queue ---

```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	ODELIST(Reason)
39361	defq	fit3143_ esoo0013	R		0:54	2	student-caas-n[01-02]

```

--- latest log ---
Building...
Build OK

FIT3143 Lab 2, CAAS results
Erwyna Soo Wen Xin (36555789), Taabish Farooq Bhat (35473932)
Job 39361, Wed Sep 9 17:40:16 AEST 2026
n = 130000000
Nodes allocated: 2 (student-caas-n[01-02])
Tasks: 8, per node 4, cpus per task 2
=====

--- 1. Serial baseline (Week 4 Task 1) ---
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault %

```

27. Prompt, Wednesday 9 September 2026, 3:41 PM

```
scancel 39360
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % scancel
39360
zsh: command not found: scancel
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % cd ~/Week8_Lab/caas
sed -i 's|\\.\./\\.\./Lab1_W4/task1\\.c|../serial_instr.c|; s|primes_serial\\.txt|primes_serial_instr.txt|'
run_all.job
cd: no such file or directory: /Users/erwynasoo/Week8_Lab/caas
sed: -I or -i may not be used with stdin
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % grep -n "caas_serial\\|primes_serial" run_all.job
grep: run_all.job: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % grep -n
"caas_serial\\|primes_serial" run_all.job
grep: run_all.job: No such file or directory
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % sbatch run_all.job
squeue -u esoo0013
zsh: command not found: sbatch
zsh: command not found: squeue
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % squeue -u esoo0013
zsh: command not found: squeue
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
0013@student-caas-headnode.rep.monash.edu 'scancel 3936
0 2>/dev/null; cd ~/Week8_Lab/caas && sed -i "s|\\.\./\\.\
./Lab1_W4/task1\\.c|../serial_instr.c|; s|primes_serial
\\.txt|primes_serial_instr.txt|" run_all.job && echo "---
- patched lines ---" && grep -n "caas_serial\\|primes_se
rial" run_all.job && echo "--- submitting ---" && sbatc
h run_all.job && sleep 20 && echo "--- queue after 20s
---" && squeue -u esoo0013'
esoo0013@student-caas-headnode.rep.monash.edu's password:
--- patched lines ---
45:gcc -O2 ../serial_instr.c -o caas_serial -lm || { echo "serial build FAILED"; exit 1; }
62: ./caas_serial $N
84: tail -n +2 primes_serial_instr.txt > ref.txt 2>/dev/null
--- submitting ---
Submitted batch job 39361
--- queue after 20s ---
      JOBID PARTITION    NAME        USER ST       TIME  NODES NODELIST(REASON)
      39361      defq fit3143_ esoo0013  R        0:19      2 student-caas-n[01-02]
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo0013@student-caas-headnode.rep.monash.edu 'echo "---
queue ---"; squeue -u esoo0013; echo "--- latest log ---"; cd ~/Week8_Lab/caas && ls -t lab2_caas*.out | head
-1 | xargs tail -30'
esoo0013@student-caas-headnode.rep.monash.edu's password:
--- queue ---
      JOBID PARTITION    NAME        USER ST       TIME  NODES NODELIST(REASON)
      39361      defq fit3143_ esoo0013  R        0:54      2 student-caas-n[01-02]
--- latest log ---
Building...
Build OK

FIT3143 Lab 2, CAAS results
Erwyna Soo Wen Xin (36555789), Taabish Farooq Bhat (35473932)
Job 39361, Wed Sep  9 17:40:16 AEST 2026
n = 130000000
Nodes allocated: 2 (student-caas-n[01-02])
Tasks: 8, per node 4, cpus per task 2
=====

--- 1. Serial baseline (Week 4 Task 1) ---
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo0013@student-caas-headnode.rep.monash.edu 'squeue -u
esoo0013; echo "---"; tail -40 ~/Week8_Lab/caas/caas_results.txt 2>/dev/null'
esoo0013@student-caas-headnode.rep.monash.edu's password:
      JOBID PARTITION    NAME        USER ST       TIME  NODES NODELIST(REASON)
      39361      defq fit3143_ esoo0013  R        1:35      2 student-caas-n[01-02]
---
FIT3143 Lab 2, CAAS results
Erwyna Soo Wen Xin (36555789), Taabish Farooq Bhat (35473932)
Job 39361, Wed Sep  9 17:40:16 AEST 2026
n = 130000000
Nodes allocated: 2 (student-caas-n[01-02])
Tasks: 8, per node 4, cpus per task 2
=====

--- 1. Serial baseline (Week 4 Task 1) ---
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault %
```


am I done now

28. Prompt, Wednesday 9 September 2026, 3:48 PM

```
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault % ssh esoo
0013@student-caas-headnode.rep.monash.edu 'squeue -u es
oo0013; echo "=== RESULTS ==="; cat ~/Week8_Lab/caas/ca
as_results.txt'
esoo0013@student-caas-headnode.rep.monash.edu's password:
      JOBID PARTITION      NAME      USER ST      TIME  NODES NODELIST(REASON)
=== RESULTS ===
FIT3143 Lab 2, CAAS results
Erwyna Soo Wen Xin (36555789), Taabish Farooq Bhat (35473932)
Job 39361, Wed Sep  9 17:40:16 AEST 2026
n = 130000000
Nodes allocated: 2 (student-caas-n[01-02])
Tasks: 8, per node 4, cpus per task 2
=====

--- 1. Serial baseline (Week 4 Task 1) ---
PHASE,130000000,1,1,7378187,77.600156,0.693158,78.293314

--- 2. Open MPI (Task 1) across nodes ---
[ p = 1 ]
Task 1 (Open MPI) Summary:
  Input n: 130000000
  Total Primes Found: 7378187
  Computation Time: 51.5379 seconds
  Total Wall Clock Time: 52.4706 seconds

[ p = 2 ]
Task 1 (Open MPI) Summary:
  Input n: 130000000
  Total Primes Found: 7378187
  Computation Time: 25.7815 seconds
  Total Wall Clock Time: 26.8378 seconds

[ p = 4 ]
Task 1 (Open MPI) Summary:
  Input n: 130000000
  Total Primes Found: 7378187
  Computation Time: 12.8873 seconds
  Total Wall Clock Time: 13.9890 seconds

[ p = 8 ]
Task 1 (Open MPI) Summary:
  Input n: 130000000
  Total Primes Found: 7378187
  Computation Time: 6.4443 seconds
  Total Wall Clock Time: 7.5934 seconds

--- 3. Open MPI, per phase, to measure the gather across the network ---
PHASE line format: n,procs,threads,primes,bcast,comp,imbal,gather,sort,write,total,comp_min,comp_mean
[ p = 1 ]
PHASE,130000000,1,1,7378187,0.000001,51.547480,0.000001,0.007923,0.298390,0.640025,52.495422,51.547480,51.547480

[ p = 2 ]
PHASE,130000000,2,1,7378187,0.000020,25.836796,0.065091,0.037984,0.352047,0.636467,26.864777,25.771939,25.804367

[ p = 4 ]
PHASE,130000000,4,1,7378187,0.009813,12.912668,0.034245,0.050938,0.397093,0.618169,13.989801,12.884690,12.899586

[ p = 8 ]
PHASE,130000000,8,1,7378187,0.001524,6.454557,0.011457,0.063339,0.432473,0.642648,7.594095,6.443369,6.446300

--- 4. Hybrid (Task 2), one process per node ---
OMP_NUM_THREADS = 2
Task 2 (Hybrid Open MPI + OpenMP) Summary:
  Input n: 130000000
  MPI Processes: 2 | OMP Threads per Process: 2
  Total Primes Found: 7378187
  Computation Time: 12.9095 seconds
  Total Wall Clock Time: 13.9627 seconds
=====
Correctness check against the serial reference:
Task 1 output MATCHES the serial reference
erwynasoo@Erwynas-MacBook-Pro Obsidian Vault %
```

does this mena I am done

29. Prompt, Wednesday 9 September 2026, 3:53 PM

Make everything submission ready to submit remove the draft slides I will add the slides there myself, do they want me to submit all in one zipped file or like seperately with slides pdf seperate from code and ai declaration too... how???

Make sure comments are in my voice / taabish's voice, taabish voice for task 1 and 2 and mine for t 2 and 3 ya do not put there erwyna when its a taabish task yea! Make sure comments sound in our voices and not ai or robotic, over-explaining or hyphens

<https://canva.link/r2mw03o8q0bmo5f>

it should allows edits this link!

30. Prompt, Wednesday 9 September 2026, 3:54 PM

Make everything submission ready to submit remove the draft slides I will add the slides there myself, do they want me to submit all in one zipped file or like seperately with slides pdf seperate from code and ai declaration too... how???

Make sure comments are in my voice / taabish's voice, taabish voice for task 1 and 2 and mine for t 2 and 3 ya do not put there erwyna when its a taabish task yea! Make sure comments sound in our voices and not ai or robotic, over-explaining or hyphens

<https://canva.link/r2mw03o8q0bmo5f>

it should allows edits this link!

And return a zipped, submission ready folder with everything they do not want liek txt files removed

31. Prompt, Wednesday 9 September 2026, 4:08 PM

Do I really need /Users/erwynasoo/FIT3143/FIT3143_Lab2_Submission/Task3_Performance_Evaluation.pdf and /Users/erwynasoo/FIT3143/FIT3143_Lab2_Submission/SUBMISSION_NOTES.txt isn't that overkill and again should everythiong be in one zip or seperate like the most recent applied task

32. Prompt, Wednesday 9 September 2026, 4:25 PM

<https://canva.link/r2mw03o8q0bmo5f>

do the changes directly here

33. Prompt, Wednesday 9 September 2026, 4:28 PM

change my pdf to say gemini flash

Session 2, 15 and 16 September 2026: final check against the specification and rubric, per rank timing, the Task 3 sweep over n, and Q&A practice

34. Prompt, Tuesday 15 September 2026, 9:46 PM

All right, so actually for parallel computing, I got my results from lab one and I got quite a disappointment for my result. Um, I gave you my feedback previously. I think that saved in a vault. If not, please look at that conversation. It's called lab one, three, one, four, three. Um, I wrote an email to my teacher basically asking if they could have a possible remark for me since I did do what they asked, but they did remove my marks, exceptionally for even the spoken interview. So I do need you to like help me answer exactly what they want because I think I was not as specific as I should be. For my 3143 lab one interview uh, answers. So now for lab two, I want you to go into my VSC code, like which I provided to you before and which you have worked in previously and just make sure that everything is in tip top condition. And that everything supports the lab two um, requirements uh, nothing goes against the requirements. All to get a HD and making sure triple check, double check that everything is submissibile in a way that, okay, Tabish did this task, so Tabish should be the one commenting, or Weena did this task, I did this task, so it should be in my voice, etc., etc. And nothing seems very robotic or AI. Uh, in a way that's like very hard to understand or overcomplicated and make sure that you stick exactly to what they want and what they said in the class transcripts in the slides everything in very very much detail because you saw for lab one I actually used Fable but it went so terribly wrong I got a D instead of a HD which is horrendous and I don't want that to repeat that's why right now I'm asking you to actually re-look at everything in the vault to really make sure that I've completed everything I don't have any suspicious things like erwina.md file that has my personal notes in it etc etc

35. Answer to Claude's questions, Tuesday 15 September 2026, 10:36 PM

Question: How far should I go on task1.c and task2.c (Taabish's code) before you submit?

Answer: Tidy + per-rank timing (Recommended)

Question: Besides Claude, did you or Taabish use any other AI tool (Gemini, ChatGPT, Copilot, etc.) for any part of Lab 2?

Answer: Another tool was used

Question: When is your Lab 2 class? The Moodle deadline is the start of that class.

Answer: Thu 17 Sep

36. Answer to Claude's questions, Tuesday 15 September 2026, 10:52 PM

Question: Which other AI tool was used for Lab 2, and by whom? Tick all that apply, and type what it was used for in the notes (e.g. 'first draft of task1.c').

Answer: Gemini, by Taabish

Question: Spec item 9 says ALL prompt records must be uploaded as PDF. Can whoever used that tool export those chats before Thursday?

Answer: i will include this later concentrate on the HD reqs for now

37. Prompt, Wednesday 16 September 2026, 2:45 PM

<https://canva.link/r2mw03o8q0bmo5f>

update the slides for me

taabish accidentally worked on slide 1-5 in here <https://canva.link/mbdkdu4qsekq68a>, can Check if any of the information is already in the actual slides and update if not for the actual slides. Replace the wording if wrong. Make sure everything is in CAAS and follows lab one instructions and data.

38. Prompt, Wednesday 16 September 2026, 3:37 PM

@"/Users/erwynasoo/Desktop/Hybrid MPI and OpenMP Programming.pdf"

this is My and Tabish's work I need you to renew the AI declaration to follow this instead of um, using the previous AI declaration. This is the new one with the accurate information and exact prompting that I use and Tavish use. Please be very, very specific. And make sure that you completely replace the current one with this one. He used a Gemini flash, right? Gemini yes. flash. Gemini Pro. Gemini Pro. Do not say that you use Claude. We did not use Claude. We used Gemini Pro, for everything. So just make sure that with all the lines line up withw hat is on this doc. I need you to renew the AI declaration to follow this instead of um, using the previous AI declaration. This is the new one with the accurate information and exact prompting that I use and Tabish use. Please be very, very specific. And make sure that you completely replace the current one with this one. He used a uh, Gemini Flash, right? Gemini yes. Flash. Gemini Pro. Gemini Pro. Do not say that you used Claude. We did not use Claude. We used Gemini Pro for everything. So just make sure that with all the lines. Uh, MacBook Pro 14 inch Nov 2023 not a wiondows you are opening it in aaron's laptop my email is erwynasoo@gmail.com and the canva linksa re open on Y LAPTOP

The above commands failed pls do it on MY laptop

<https://www.canva.com/design/DAHUmDDuFms/L2zbrvbCY7BqgKbzuwSfBA/edit>

https://www.canva.com/design/DAHUmeQyVKE/4wJ-nUce7tNRvMv6hK_Bqw/edit

39. Prompt, Wednesday 16 September 2026, 3:42 PM

ok fine do both tools then finish the canva edits and look at /Users/erwynasoo/FIT3143/Week8_Lab/graphs lmk if any graphs I need to update

40. Prompt, Wednesday 16 September 2026, 3:50 PM

This session is being continued from a previous conversation that ran out of context. The summary below covers the earlier portion of the conversation.

Summary:

1. **Primary Request and Intent:**

- **(Message 1)** Erwyna (FIT3143 Parallel Computing, Monash Malaysia, S2 2026) got a disappointing Lab 1 result (71.4/100), emailed tutor Vishnu Monn asking about a remark, and wants help to "answer exactly what they want" in the lab interview/Q&A because she "was not as specific as I should be". For Lab 2 she wants her VS Code project checked so everything is "in tip top condition", supports the Lab 2 requirements, nothing goes against them, aimed at HD, "triple check, double check that everything is submissible" with correct voice attribution ("Tabish did this task, so Tabish should be the one commenting... I did this task, so it should be in my voice"), nothing "robotic or AI", not overc

[... 13,148 characters of pasted course material left out here. The full prompt was 14,448 characters long ...]

t: /Users/erwynasoo/.claude/projects/-Users-erwynasoo-Desktop-Obsidian-Vault/9d76bc22-df48-443c-8ff7-8dc3b695fed0.jsonl

Continue the conversation from where it left off without asking the user any further questions. Resume directly – do not acknowledge the summary, do not recap what was happening, do not preface with "I'll continue" or similar. Pick up the last task as if the break never happened.

Session 3, 16 September 2026: last check of the finished slides and submission files, and putting the AI declaration and both prompt records into one file

41. Prompt, Wednesday 16 September 2026, 5:38 PM

19 (A1) graph8_phase_breakdown.png subtitle said task1_instr.c
23 (A5) graph10_amdahl_vs_n.png new slide, empty spot waiting

for 3 and 8 I updated the graphs but for 19 and 23 not sure what to do what ss do i put for 23 i just put graph 23

do i NEED Task3_Performance_Evaluation.pdf · or not necessary make sure all is submission ready

<https://canva.link/r2mw03o8q0bmo5f>

And is everything from my old slides <https://canva.link/mbdkdu4qsekq68a>

in the new one above:

/Users/erwynasoo/FIT3143/Week8_Lab

make appriopraite edist for truth sake i used gemini fully with taabish so theres one AI declaration make sure all meets the HD requirements and update the vault, is everything all submission ready HD ready and not lsoe makr sliiek i did for AI

42. Prompt, Wednesday 16 September 2026, 5:58 PM

@"/Users/erwynasoo/Desktop/AI_Declaration copy.pdf"
/Users/erwynasoo/Desktop/FIT3143_Lab2_Submission

Is this submission ready?

compare with my 3143 vault

do NOT make a new AI declaration, only use THIS the actual and can you remake
/Users/erwynasoo/FIT3143/Week8_Lab/Task3_Performance_Evaluation.md into a pdf

43. Prompt, Wednesday 16 September 2026, 6:09 PM

```
<bash-input>cd ~/Desktop && rm -f FIT3143_Lab2_Submission.zip && zip -r -X FIT3143_Lab2_Submission.zip  
FIT3143_Lab2_Submission -x "*.DS_Store"</bash-input><bash-stdout>  adding: FIT3143_Lab2_Submission/ (stored 0%)  
  adding: FIT3143_Lab2_Submission/README.md (deflated 47%)  
  adding: FIT3143_Lab2_Submission/AI_Declaration.pdf (deflated 26%)  
  adding: FIT3143_Lab2_Submission/task2.c (deflated 67%)  
  adding: FIT3143_Lab2_Submission/task1.c (deflated 64%)  
  adding: FIT3143_Lab2_Submission/supporting/ (stored 0%)  
  adding: FIT3143_Lab2_Submission/supporting/caas/ (stored 0%)  
  adding: FIT3143_Lab2_Submission/supporting/caas/caas_results.txt (deflated 50%)  
  adding: FIT3143_Lab2_Submission/supporting/caas/run_all.job (deflated 57%)  
  adding: FIT3143_Lab2_Submission/supporting/caas/hybrid.job (deflated 46%)  
  adding: FIT3143_Lab2_Submission/supporting/caas/README.md (deflated 46%)  
  adding: FIT3143_Lab2_Submission/supporting/caas/serial.job (deflated 46%)  
  adding: FIT3143_Lab2_Submission/supporting/caas/CAAS_Analysis.pdf (deflated 5%)  
  adding: FIT3143_Lab2_Submission/supporting/caas/mpi.job (deflated 46%)  
  adding: FIT3143_Lab2_Submission/supporting/results/ (stored 0%)  
  adding: FIT3143_Lab2_Submission/supporting/results/amdahl_by_n.csv (deflated 66%)  
  adding: FIT3143_Lab2_Submission/supporting/results/results_by_procs.csv (deflated 56%)  
  adding: FIT3143_Lab2_Submission/supporting/results/phases_by_n_task2.csv (deflated 63%)  
  adding: FIT3143_Lab2_Submission/supporting/results/phases_by_n_task1.csv (deflated 63%)  
  adding: FIT3143_Lab2_Submission/supporting/results/phases_serial.csv (deflated 32%)  
  adding: FIT3143_Lab2_Submission/supporting/results/results_hybrid_t.csv (deflated 45%)  
  adding: FIT3143_Lab2_Submission/supporting/results/results_hybrid_total.csv (deflated 46%)  
  adding: FIT3143_Lab2_Submission/supporting/results/rank_balance.csv (deflated 54%)  
  adding: FIT3143_Lab2_Submission/supporting/results/amdahl_summary.csv (deflated 63%)  
  adding: FIT3143_Lab2_Submission/supporting/results/results_by_n.csv (deflated 54%)  
  adding: FIT3143_Lab2_Submission/supporting/results/phases_task2.csv (deflated 60%)  
  adding: FIT3143_Lab2_Submission/supporting/results/phases_task1.csv (deflated 60%)  
  adding: FIT3143_Lab2_Submission/supporting/instruments/ (stored 0%)  
  adding: FIT3143_Lab2_Submission/supporting/instruments/task2_instr.c (deflated 68%)  
  adding: FIT3143_Lab2_Submission/supporting/instruments/serial_instr.c (deflated 60%)  
  adding: FIT3143_Lab2_Submission/supporting/instruments/task1_instr.c (deflated 65%)  
  adding: FIT3143_Lab2_Submission/supporting/scripts/ (stored 0%)  
  adding: FIT3143_Lab2_Submission/supporting/scripts/run_benchmarks.sh (deflated 66%)  
  adding: FIT3143_Lab2_Submission/supporting/scripts/run_phases_by_n.py (deflated 57%)  
  adding: FIT3143_Lab2_Submission/supporting/scripts/run_phases.py (deflated 58%)  
  adding: FIT3143_Lab2_Submission/supporting/scripts/make_diagrams.py (deflated 67%)  
  adding: FIT3143_Lab2_Submission/supporting/scripts/make_graphs.py (deflated 70%)  
  adding: FIT3143_Lab2_Submission/supporting/scripts/measure_launch.py (deflated 50%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/ (stored 0%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/diagram1_stride_partition.png (deflated 12%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/graph3_speedup_vs_width.png (deflated 11%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/graph2_speedup_vs_n.png (deflated 8%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/diagram3_hybrid_layout.png (deflated 12%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/graph5_matched_width.png (deflated 9%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/graph6_amdahl_task1.png (deflated 8%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/diagram2_gatherv_offsets.png (deflated 10%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/graph1_runtime_vs_n.png (deflated 10%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/graph8_phase_breakdown.png (deflated 18%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/graph7_amdahl_task2.png (deflated 12%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/graph4_hybrid_vs_mpi_threads.png (deflated 15%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/graph9_rank_balance.png (deflated 16%)  
  adding: FIT3143_Lab2_Submission/supporting/graphs/graph10_amdahl_vs_n.png (deflated 9%)
```

```
adding: FIT3143_Lab2_Submission/Task3_Performance_Evaluation.pdf (deflated 21%)</bash-stdout><bash-stderr></bash-stderr>
```

44. Prompt, Wednesday 16 September 2026, 9:55 PM

```
@"/Users/erwynasoo/Downloads/FIT3143 2026 Lab #2 Consultation transcript_2026-09-16_15.11.51.txt"
this was teh consultation about lab 2 see if i respect all in the zip and folder
```

45. Prompt, Wednesday 16 September 2026, 10:01 PM

```
<bash-input>cd ~/FIT3143/Week8_Lab && python3 make_pdfs.py Task3_Performance_Evaluation.md
Task3_Performance_Evaluation.pdf && cp Task3_Performance_Evaluation.pdf ~/Desktop/FIT3143_Lab2_Submission/ && cd
~/Desktop && rm -f FIT3143_Lab2_Submission.zip && zip -r -X FIT3143_Lab2_Submission.zip FIT3143_Lab2_Submission
-x "*.DS_Store"</bash-input><bash-stdout>wrote
/Users/erwynasoo/FIT3143/Week8_Lab/Task3_Performance_Evaluation.pdf
adding: FIT3143_Lab2_Submission/ (stored 0%)
adding: FIT3143_Lab2_Submission/README.md (deflated 48%)
adding: FIT3143_Lab2_Submission/AI_Declaration.pdf (deflated 24%)
adding: FIT3143_Lab2_Submission/task2.c (deflated 67%)
adding: FIT3143_Lab2_Submission/task1.c (deflated 64%)
adding: FIT3143_Lab2_Submission/supporting/ (stored 0%)
adding: FIT3143_Lab2_Submission/supporting/caas/ (stored 0%)
adding: FIT3143_Lab2_Submission/supporting/caas/caas_results.txt (deflated 50%)
adding: FIT3143_Lab2_Submission/supporting/caas/run_all.job (deflated 57%)
adding: FIT3143_Lab2_Submission/supporting/caas/hybrid.job (deflated 46%)
adding: FIT3143_Lab2_Submission/supporting/caas/README.md (deflated 46%)
adding: FIT3143_Lab2_Submission/supporting/caas/serial.job (deflated 46%)
adding: FIT3143_Lab2_Submission/supporting/caas/CAAS_Analysis.pdf (deflated 5%)
adding: FIT3143_Lab2_Submission/supporting/caas/mpi.job (deflated 46%)
adding: FIT3143_Lab2_Submission/supporting/results/ (stored 0%)
adding: FIT3143_Lab2_Submission/supporting/results/amdahl_by_n.csv (deflated 66%)
adding: FIT3143_Lab2_Submission/supporting/results/results_by_procs.csv (deflated 56%)
adding: FIT3143_Lab2_Submission/supporting/results/phases_by_n_task2.csv (deflated 63%)
adding: FIT3143_Lab2_Submission/supporting/results/phases_by_n_task1.csv (deflated 63%)
adding: FIT3143_Lab2_Submission/supporting/results/phases_serial.csv (deflated 32%)
adding: FIT3143_Lab2_Submission/supporting/results/results_hybrid_t.csv (deflated 45%)
adding: FIT3143_Lab2_Submission/supporting/results/results_hybrid_total.csv (deflated 46%)
adding: FIT3143_Lab2_Submission/supporting/results/rank_balance.csv (deflated 54%)
adding: FIT3143_Lab2_Submission/supporting/results/amdahl_summary.csv (deflated 63%)
adding: FIT3143_Lab2_Submission/supporting/results/results_by_n.csv (deflated 54%)
adding: FIT3143_Lab2_Submission/supporting/results/results/phases_task2.csv (deflated 60%)
adding: FIT3143_Lab2_Submission/supporting/results/phases_task1.csv (deflated 60%)
adding: FIT3143_Lab2_Submission/supporting/instruments/ (stored 0%)
adding: FIT3143_Lab2_Submission/supporting/instruments/task2_instr.c (deflated 68%)
adding: FIT3143_Lab2_Submission/supporting/instruments/serial_instr.c (deflated 60%)
adding: FIT3143_Lab2_Submission/supporting/instruments/task1_instr.c (deflated 65%)
adding: FIT3143_Lab2_Submission/supporting/scripts/ (stored 0%)
adding: FIT3143_Lab2_Submission/supporting/scripts/run_benchmarks.sh (deflated 66%)
adding: FIT3143_Lab2_Submission/supporting/scripts/run_phases_by_n.py (deflated 57%)
adding: FIT3143_Lab2_Submission/supporting/scripts/run_phases.py (deflated 58%)
adding: FIT3143_Lab2_Submission/supporting/scripts/make_diagrams.py (deflated 67%)
adding: FIT3143_Lab2_Submission/supporting/scripts/make_graphs.py (deflated 70%)
adding: FIT3143_Lab2_Submission/supporting/scripts/measure_launch.py (deflated 50%)
adding: FIT3143_Lab2_Submission/supporting/graphs/ (stored 0%)
adding: FIT3143_Lab2_Submission/supporting/graphs/diagram1_stride_partition.png (deflated 12%)
adding: FIT3143_Lab2_Submission/supporting/graphs/graph3_speedup_vs_width.png (deflated 11%)
adding: FIT3143_Lab2_Submission/supporting/graphs/graph2_speedup_vs_n.png (deflated 8%)
adding: FIT3143_Lab2_Submission/supporting/graphs/diagram3_hybrid_layout.png (deflated 12%)
adding: FIT3143_Lab2_Submission/supporting/graphs/graph5_matched_width.png (deflated 9%)
adding: FIT3143_Lab2_Submission/supporting/graphs/graph6_amdahl_task1.png (deflated 8%)
adding: FIT3143_Lab2_Submission/supporting/graphs/diagram2_gatherv_offsets.png (deflated 10%)
adding: FIT3143_Lab2_Submission/supporting/graphs/graph1_runtime_vs_n.png (deflated 10%)
adding: FIT3143_Lab2_Submission/supporting/graphs/graph8_phase_breakdown.png (deflated 18%)
adding: FIT3143_Lab2_Submission/supporting/graphs/graph7_amdahl_task2.png (deflated 12%)
adding: FIT3143_Lab2_Submission/supporting/graphs/graph4_hybrid_vs_mpi_threads.png (deflated 15%)
adding: FIT3143_Lab2_Submission/supporting/graphs/graph9_rank_balance.png (deflated 16%)
adding: FIT3143_Lab2_Submission/supporting/graphs/graph10_amdahl_vs_n.png (deflated 9%)
adding: FIT3143_Lab2_Submission/Task3_Performance_Evaluation.pdf (deflated 21%)</bash-stdout><bash-stderr></bash-stderr>
```

46. Prompt, Wednesday 16 September 2026, 10:05 PM

```
https://canva.link/r2mw03o8q0bmo5f add the script in notes
```